

# ARTIFICIAL INTELLIGENCE AND SPONTANEOUS COLLUSION\*

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## Abstract

We develop a tractable model for studying strategic interactions between learning algorithms. We uncover a mechanism, responsible for the emergence of algorithmic collusion, that enables algorithms to periodically coordinate on actions that are more profitable than static Nash equilibria. This novel collusive channel, spontaneous coupling, relies on an endogenous statistical linkage in the algorithms' estimates. The model's parameters predict whether the statistical linkage will appear, and what market structures facilitate algorithmic collusion. We show that spontaneous coupling can sustain collusion in prices and market shares, complementing numerical findings in the literature. Finally, we apply our results to design algorithmic markets.

**Keywords:** Artificial Intelligence, Learning, Collusion

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# 1 Introduction

Artificial Intelligence (AI) software is becoming more common in a variety of business contexts, ranging from bidding in online auctions to pricing on shopping platforms and rent setting. This market shift has raised concerns that automated pricing and bidding software could facilitate collusive behavior, voiced both by regulatory authorities (see [OECD, 2017](#); [Competition Bureau, 2018](#); [Competition & Markets Authority, 2021](#)) and by academic research (see [Harrington, 2018](#); [Calvano et al., 2020](#); [Asker, Fershtman and Pakes, 2022](#)).

In this paper, we identify a novel channel that facilitates seemingly collusive behavior between AI algorithms. Unlike explicit cartels or tacitly-collusive equilibria, this channel does not require deliberate intent to collude. Instead, supra-competitive outcomes arise endogenously as a result of a statistical linkage between independent, myopic, profit-maximizing learning algorithms. To highlight these characteristics, we call this channel *spontaneous coupling*. We show that spontaneous coupling hinges on the algorithmic nature of market participants, rather than on their monitoring technology and strategies.

A major challenge in analyzing games played by learning algorithms is that the evolution of play is stochastic and discrete. Since AI algorithms learn by running randomized experiments, they generate stochastic, non-stationary paths of play. To handle these difficulties, we approximate these systems in continuous time. This allows us to leverage dynamical systems techniques to characterize analytically the algorithms' play.

We first develop our insights in a Prisoner's Dilemma. Despite the strategic simplicity of the game, AI algorithms such as naive Q-learning (not designed to learn dynamic reward strategies such as tit-for-tat) may enter stochastic cycles that exhibit high cooperation rates. The mechanism we isolate is responsible for such cooperation. The algorithms we consider estimate payoffs from each action by running experiments, and exploit their estimates to chase larger rewards. When experiments are infrequent, the estimates persist for longer periods of play, eventually becoming outdated and thus biased. Such biases are correlated across algorithms, which synchronizes their paths of play: agents act symmetrically, jointly cooperating and defecting in cycles of random length. Moreover, our analysis shows that these cycles become longer the more attractive cooperation is. We call this phenomenon *spontaneous coupling*. On the other hand, we prove that coupling disappears when experiments are frequent enough or when there are numerous participants, as spontaneously correlated estimates become less likely.

We then show how these insights can be generalized and applied to more diverse strategic settings. First, we show both theoretically and in simulations that spontaneous

coupling may arise between *reinforcers*, a class of algorithms that includes many reinforcement learning procedures. Through simulations, we show that spontaneous coupling also appears in a more complex game, sustaining collusive behavior such as algorithmic market splitting. Second, by isolating the features which lead to spontaneous coupling, we characterize a class of algorithms that are immune to it: procedures in this class are designed to incorporate careful exploration or counterfactual modeling, and are guaranteed to learn to play undominated strategies in any finite game. This characterization allows us to make sense of the results of [Asker, Fershtman and Pakes \(2022\)](#): spontaneous coupling sustains the collusive outcomes they observe, and their “synchronous” algorithms (which eliminate collusion) fall in the class we characterize as immune to coupling.

Finally, we propose a market intervention to prevent spontaneous coupling. Motivated by online auctions and marketplaces, we take the perspective of a market designer. We show that it is possible to design strategy-proof mechanisms that are robust to the participation of algorithmic players. Specifically, we prove that the statistical linkage we identify disappears if the designer provides enough feedback to the algorithms to assist their learning, and we characterize feedback policies with minimal communication.

## 1.1 Intuition

Before introducing the formal model, we sketch the main intuition of the paper and how it generates the main results.

In [Section 2](#) we introduce a model of algorithmic learning and the mathematical formalism to analyze its dynamics; we call *reinforcers* algorithms that fit this model. Algorithms in this class maintain a vector of values representing each action’s payoff consequences. That is, the algorithm assigns to each action a *perception*, a subjective estimate of the reward of playing that action. A reinforcer estimates his perceptions by repeatedly interacting with the environment and updating them based on the observed payoff.<sup>1</sup> A policy function is responsible for trading off exploration (running experiments, to estimate payoffs accurately) and exploitation (selecting what is thought to be the optimal action, to collect rewards). For example, an  $\varepsilon$ -greedy policy selects the action with the currently highest perception with probability  $1-\varepsilon$ , and with probability  $\varepsilon$  explores the action space uniformly at random. Algorithms in this class form the bedrock of currently

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<sup>1</sup>For example, this class includes the celebrated Q-learning procedure, itself the building block of many AI algorithms, as well as some variants of the Multiplicative Weights Update. However, it does not include algorithms such as (stochastic) Gradient Ascent, which optimize over the space of policies directly, or Large Language Models (LLMs), which are trained *before* being deployed.

deployed unsupervised learning and reinforcement learning procedures.<sup>2</sup>

In [Section 3](#) we pin down the mechanism that sustains cooperation (see below for an intuitive summary). In practice, algorithms play symmetric profiles of actions “too often”. We dub this phenomenon *spontaneous* coupling to stress that even algorithms exploring independently may get linked. We prove ([Proposition 1](#)) that coupling leads to the continuous-time counterpart of cooperation cycles, where agents cooperate most but not all the time.<sup>3</sup>

Intuitively, during a period of collusion, each agent estimates the value of colluding *conditional on the opponent colluding (most of the time) as well*. This increases the value of the perception of collusion. At the same time, since experimenting with the competitive action yields an even higher flow profit, exploration leads the agents to estimate a high payoff for competition. However, as soon as one agent begins exploiting the competitive action, the opponent will quickly best-respond by competing too, and joint competition will yield reduced payoffs. In particular, the perception of the competitive action now decreases: the agents estimate the value of competing *conditional on the opponent competing as well*. Instead, because the agent explores seldom, the perception of collusion remains close to the value of collusion *conditional on the opponent colluding as well*. This draws the algorithms away from competition and back into a cycle of collusive behavior. This is how spontaneous coupling sustains dominated outcomes: simultaneous deviations reinforce the estimation error in the algorithms’ perceptions.

In [Section 4](#) we show ([Theorem 2](#)) that coupling relies on a set of parameters, the relative learning rates of an algorithm, i.e., the speed at which each action’s estimate gets updated over time. The slower an algorithm learns about infrequently played actions, the more persistent their perceptions. Persistence, combined with nearly simultaneous deviations, leads to sustained periods of correlated play and ultimately, collusion. This form of coordination differs from both repeated-game strategies and one-shot coordination through an intermediary (correlated and coarse-correlated equilibria), all of which rest on some form of rationality. Spontaneous coupling is not specific to one class of games because of its high-level correspondence to coarse-correlated equilibria, but because it relies on the algorithmic nature of the players, it enables coordination even where equilibrium theory deems it impossible. On the other hand, algorithms with uniform learning rates,

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<sup>2</sup>While precise estimates are impossible to obtain because of privacy concerns, surveys report that about 25% of pricing managers delegate dynamic pricing decision to Q-learning or unsupervised learning algorithm, and nearly half of them do not employ their competitors’ prices in the algorithms they use ([Spann et al. \(2024\)](#)). Procedures like these precisely fall in the class we consider.

<sup>3</sup>This explains why we observe that, while such algorithms often learn to collude in simulations, collusion is imperfect. It is because the agents cannot converge to a constant profile of actions that is not a pure Nash Equilibrium — these AIs always learn how to best respond to a fixed profile of actions.

where each action’s perception is updated at the same speed, avoid the stochastic cycles and learn to play only undominated strategies, because uniform learning rates guarantee that every action enjoys the same persistence ([Theorems 3 and 4](#)).

## 1.2 Literature Review

The literature on algorithmic collusion is growing through both experimental work (see, e.g., [Klein, 2021](#); [Abada and Lambin, 2023](#); [Johnson, Rhodes and Wildenbeest, 2023](#)) and empirical work (see, e.g., [Musolff, 2021](#); [Assad et al., 2023](#)). The seminal results of [Calvano et al. \(2020\)](#) focus on strategies as a proxy for collusion, and show that algorithms learn reward-punishment strategies. By removing monitoring and conditioning, we show instead how an algorithm’s design can systematically lead to collusive outcomes even absent traditionally understood motives or tools. Our results shed light on the experimental results of [Asker, Fershtman and Pakes \(2022\)](#) in a Bertrand oligopoly setting, and [Banchio and Skrzypacz \(2022\)](#) in various auction settings. Much like [Johnson, Rhodes and Wildenbeest \(2023\)](#), we study how a platform can limit the scope of algorithmic collusion with market design tools.

Unlike much theoretical work on algorithmic collusion (see, e.g., [Brown and MacKay, 2023](#); [Leisten, 2022](#); [Harrington, 2022](#); [Lamba and Zhuk, 2022](#)), we explicitly model the effect of *learning* on market outcomes, in an attempt to represent the current state of AI pricing and bidding intelligence. Closer in spirit are the models of [Hansen, Misra and Pai \(2021\)](#) and [Possnig \(2023\)](#). Similarly to [Hansen, Misra and Pai \(2021\)](#), we study algorithms with no monitoring and observe convergence on supra-competitive outcomes. Interestingly, the mechanism behind the two results is significantly different: while [Hansen, Misra and Pai \(2021\)](#) points to correlation in experimentation procedures, we identify correlation in the algorithms’ profit estimates as the driver of cooperation. [Possnig \(2023\)](#) instead develops a stochastic approximation framework to analyze algorithms with monitoring technologies, and is thus closer in spirit to [Calvano et al. \(2020\)](#), [Bertrand et al. \(2023\)](#), and the literature that followed. We study markedly different algorithmic classes, and we focus our work on simpler procedures that are often employed in practical settings.

Learning, and specifically *reinforcement* learning in games, has been studied both by economists and computer scientists, both experimentally ([Erev, Bereby-Meyer and Roth, 1999](#); [Lerer and Peysakhovich, 2017](#)) and analytically (e.g., [Börgers and Sarin, 1997](#); [Erev and Roth, 1998](#); [Mertikopoulos and Sandholm, 2016](#)). Unlike the former, we offer a systematic theoretical study of complex experimental systems. We focus on a similar class

of learning procedures as in the latter series of papers, but importantly our dynamics are non-smooth. We are not the first to show that algorithmic learners do not converge to Nash equilibrium (Hart and Mas-Colell, 2003); that dominated strategies may be played in the limit (Hofbauer and Sandholm (2011), among others); or that dynamics may be chaotic (Arneodo, Couillet and Tresser, 1980). Crucially, instead, we show that chaotic dynamics may arise even in a 2-person, 2-action symmetric game due to the non-smoothness of the learning process. More generally, we depart from the evolutionary game theory approach in that we seek to write a positive theory of algorithmic interactions, rather than providing foundations for established equilibrium concepts. Finally, we observe that a mechanism similar to the one we identify is observed in a series of papers (Karandikar et al., 1998; Bendor, Mookherjee and Ray, 2001a,b) that study aspiration-based learning. They also focus on discontinuous dynamics, interpreted as a behavioral decision-making rule. Our algorithms, while developed in the context of machine learning, exhibit similar characteristics.

Although there is substantial literature on continuous-time approximations of AI algorithms (see, e.g., Tuyls, Hoen and Vanschoenwinkel, 2005; Leonardos and Piliouras, 2022), most focus on smooth dynamics, which leads to markedly different results. Much of this literature relies on results from the stochastic approximations literature, such as Benaim (1996) and Borkar and Meyn (2000). We instead leverage results by Kurtz (1970), proving a functional law of large numbers. The benefit of this approach is an approximation to the entire process’ path, instead of its limit points alone. Both Gomes and Kowalczyk (2009) and Wunder, Littman and Babes (2010) propose a continuous-time approximation of  $\varepsilon$ -greedy Q-learning in a multi-agent setting, with our approximation being closer to the latter. As Babes, Wunder and Littman (2009), we study the dynamics of a Prisoner’s Dilemma under an assumption of symmetry; we disentangle the interplay between payoffs and exploration policies, and we address the algorithmic design questions they pose.

## 2 Model

Consider a finite normal-form game  $G = (N, (A_i)_{i \in N}, (r^i)_{i \in N})$  with  $N$  players; agents repeatedly play this game and choose actions using a learning algorithm. Let us focus on one agent, Alice, whose action set  $A_i$  has cardinality  $d_i$ ; Alice delegates decision-making to an algorithm that attempts to maximize her utility. Her utility function,  $r^i(a^i, a^{-i})$ , depends on her opponents’ actions and her own.

**Example 1.** Alice employs  $\varepsilon$ -greedy Q-learning. This algorithm consists of a vector  $Q(k) \in \mathbb{R}^{d_i}$  for every period  $k$  and a decision rule  $\pi_\varepsilon$ .

- Each entry  $Q_a(k)$  is an estimate of the long-run value after playing action  $a \in A_i$  after step  $k$ . If she plays action  $a$ , then she updates the correspondy entry  $Q_a(k+1)$  as

$$Q_a(k+1) = Q_a(k) + \alpha \left[ r(k) + \gamma \max_{a'} Q_{a'}(k) - Q_a(k) \right] \quad (1)$$

where  $r(k)$  is the payoff, the parameter  $\gamma \in [0, 1)$  is a discount factor, and  $\max_{a'} Q_{a'}(k)$  represents the continuation value: after playing action  $a$  in period  $k$ , she could play the action corresponding to the largest entry of  $Q$  forever after. Finally,  $\alpha \in (0, 1)$ , is called learning rate. Instead, the entries corresponding to actions  $a' \neq a$  are not updated.

- Given the vector  $Q(k)$ , the algorithm takes actions according to a  $\varepsilon$ -greedy policy. This decision rule,  $\pi_\varepsilon: \mathbb{R}^{d_i} \rightarrow \Delta(A_i)$ , maps each vector  $Q(k)$  to a distribution over actions:

$$\pi_\varepsilon(Q(k)) = \begin{cases} \frac{1}{|\arg\max_{a \in A_i} Q_a(k)|} & \forall a \in \arg\max_{a \in A_i} Q_a(k) \quad \text{with probability } 1 - \varepsilon \\ \frac{1}{d_i} & \forall a \in A_i \quad \text{with probability } \varepsilon \end{cases}$$

Intuitively, the  $Q$ -vector estimates the long-run value of actions by iterating over a Bellman equation. In period  $k$ , the value of an action is a convex combination of its previous estimate (weighted by  $1 - \alpha$ ) and a new Bellman estimate (weighted by  $\alpha$ ). The weight  $\alpha$  controls the persistence of the estimates, with lower values implying larger persistence of past experiences.<sup>4</sup>

The  $\varepsilon$ -greedy policy balances exploration and exploitation. With probability  $1 - \varepsilon$ , the algorithm selects the action corresponding to the highest entry of the  $Q$ -vector, which reflects the agent's belief about the best course of action at that time. In contrast, with probability  $\varepsilon$ , the algorithm takes a random action, enabling the agent to explore the action space. Because of its simplicity and attractive properties in single-agent environments, it is often used as a benchmark for more complex exploration policies.

Q-learning is a representative of a learning model that promotes successful play. We call this the *reinforcer* model.

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<sup>4</sup>The Q-learning perceptions generalize the algorithms in [Erev and Roth \(1998\)](#) and [Börger and Sarin \(1997\)](#). Note that this procedure is misspecified when used as a learning and decision rule in a game. This is because Alice's  $Q$ -vector estimates a continuation payoff  $Q_{a^i}$  as a function of Alice's own actions only, while payoffs depend also on the opponents' unobserved actions,  $a^{-i}$ . If the opponents' profile of strategies were fixed, Q-learning would converge on the best response to that profile (the proof is a simple adaptation of the arguments in [Watkins and Dayan \(1992\)](#)), but in a strategic setting, where all agents are learning, there are no such convergence guarantees.

**Definition 1.** A *reinforcer* for agent  $i$  is a pair  $(\theta^i, \pi^i)$  consisting of

- A  $d_i$ -dimensional stochastic process  $\theta^i$  over a domain  $\mathcal{T} \subset \mathbb{R}^{d_i}$  that evolves according to

$$\theta^i(k+1) = \theta^i(k) + \alpha^i D^i(a^i(k), r^i(k), \theta^i(k)),$$

where  $a^i(k) \in A_i$  is the action taken by agent  $i$  in period  $k$ ,  $\alpha^i \in \mathbb{R}_+^{d_i}$  is a vector with a learning rate for each action  $a^i \in A_i$ , and  $D^i$  is an update vector function which depends on actions  $a^i(k)$ , utilities  $r^i(k)$  and previous values  $\theta^i(k)$ .

- A *policy*  $\pi^i$ , i.e., a map  $\pi^i: \mathcal{T} \rightarrow \Delta(A_i)$ , which selects a distribution of actions for each value of the process  $\theta^i \in \mathcal{T}$ .

A reinforcer carries a statistic for each available action and selects an action in period  $k$  according to its policy. This class includes many prominent reinforcement learning and algorithmic game theory procedures (such as Q-learning and EXP3), as we describe in later sections. However, other algorithms such as (stochastic) gradient ascent (which optimizes over the policy directly), or Large Language Models (LLMs, that are extensively trained *before* being deployed and do not learn “on the fly”) are not included. Evidence of spontaneous collusion among these procedures cannot be derived from our results.

We call the process  $\theta^i$  the *perceptions* of the algorithm, to highlight that these are algorithm-specific utility estimates. Both agent  $i$ ’s and her opponents’ policies introduce randomness in the perception process  $\theta^i$ . The update function  $D^i$  depends on Alice’s realized actions directly and on her opponents’ actions through her utility. Given an initial value for  $\theta^i(0)$ , which we call the *initialization* of  $\theta^i$ , this stochastic process is well-defined. The following assumption is maintained throughout the paper.

**Assumption A1.** The functions  $D^i(a^i, r^i, \theta^i)$  and  $\pi^i(\theta)$  are Lipschitz-continuous almost-everywhere in  $\theta$ .

Let us return to [Example 1](#). The policy  $\pi_\varepsilon$  is constant on the subspace of  $\mathbb{R}^{d_i}$  where  $\operatorname{argmax}_a \theta_a^i$  is fixed and unique. The  $\operatorname{argmax}$  changes only along the lines of the form  $\{\theta^i | \theta_a^i = \theta_{a'}^i = \max \theta^i\}$ , which have zero Lebesgue measure. The discontinuities of the update function of Q lie on the same lines, and thus the update is also a.e. Lipschitz. Thus  $\varepsilon$ -greedy Q-learning satisfies [Assumption A1](#).

When multiple agents employ reinforcers, we represent the system as a single vector of dimension  $d_1 + \dots + d_N$  by stacking each agent’s perceptions, and we denote it by  $\theta(k) = (\theta^1(k), \dots, \theta^N(k))$ . Similarly,  $D$  and  $\pi$  indicate the collection of update functions and policies when missing a superscript.

## 2.1 Approximation in Continuous Time

We seek to describe the dynamics of learning of reinforcers. While relatively simple to analyze in a stationary environment, little theory exists when reinforcers learn to play against each other. For example, convergence guarantees of Q-learning rely on stationarity assumptions. Analyzing the learning path of any number of Q-learning agents requires describing discrete, stochastic updates and how these interact over time.

We deal with these difficulties using a continuous-time approach known as *fluid approximations*. We adopt the formalism first introduced by Kurtz (1970): the idea behind fluid approximations is to analyze the limit of the systems as the jumps get small and their frequency increases, which yields a typically more tractable ODE system. We expect to accurately model most online marketplaces, where decisions are taken at very high frequencies and profit estimates rely on a large number of data points.

The first step of the approximation is re-casting a reinforcer  $\theta$  as a process in continuous time, a procedure called *Poissonization*.<sup>5</sup> Formally, we consider a Poisson clock with rate  $\lambda^1 = 1$  and define  $^1\theta(t)$  as the pure-jump continuous-time process such that  $^1\theta(0) = \theta(0)$  and that is constant for all  $t$  except at the ticks  $\tau$  of the Poisson clock, when it gets updated according to  $D$ . Intuitively, one can think of a sequence of stage games that are played only at the tick of the clock. We introduced a new layer of randomness, but the path traced by  $^1\theta(t)$  remains identical to the path of the discrete  $\theta(t)$ .

We generate a sequence of processes  $(^n\theta)_{n \in \mathbb{N}}$  by increasing the rate of the Poisson clock to  $\lambda^n = n$ . The goal is to regularize the learning dynamics; therefore, we need to compensate for frequent updates by reducing the contribution of each jump to the total estimate. We do this by dividing each jump by  $\frac{1}{n}$ : in the parlance of Definition 1, at a given arrival time  $\tau$  of the Poisson process,

$$^n\theta(\tau) = ^n\theta(\tau^-) + \frac{\alpha}{n} D(a(\tau), r_\tau, ^n\theta(\tau^-)),$$

where  $^n\theta(\tau^-)$  denotes the left limit  $\lim_{t \uparrow \tau} [^n\theta(t)]$ .<sup>6</sup> Each process  $^n\theta$  has the same infinitesimal generator: the sequence  $(^n\theta)_{n \in \mathbb{N}}$  preserves the instantaneous rate of change uniformly. We prove the following result:

**Theorem 1.** *Let  $H \subset \mathcal{T}$  be such that  $D$  and  $\pi$  are Lipschitz over  $H$ , and let  $y_0 \in H$  be the*

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<sup>5</sup>Kurtz (1970) provides also a construction of the fluid approximation that does not require the Poissonization. We introduce this procedure because it makes explicit how one can move between discrete and continuous time in this setting.

<sup>6</sup>Such a limit exists because the Poisson process is càdlàg, and it represents the value of the process immediately before a jump.

initialization point of  $\theta$ . Then, the sequence of continuous-time stochastic processes  $({}^n\theta)_{n \in \mathbb{N}}$  converges in probability to the solution of the following Cauchy problem:

$$\begin{cases} \frac{d\Theta^i(t)}{dt} = \alpha \mathbb{E}_{\pi^i, \pi^{-i}} \left[ D^i(a^i, r(a^i, a^{-i}), \Theta^i(t)) \right] \\ \Theta^i(0) = y_0^i \end{cases}$$

for all  $i$ . That is,  $\lim_{n \rightarrow \infty} P \left\{ \sup_{t \leq T} \left\| {}^n\theta(t) - \Theta(t) \right\| > \eta \right\} = 0$  for all  $T \geq 0$  and  $\eta > 0$  such that  $\{\Theta(t)\}_{t \leq T} \subset H$ .

We provide a formal construction of the sequence  ${}^n\theta$  and a proof of this result in [Appendix A](#). The process  $\Theta$  is the fluid approximation to  $\theta$ : it is a deterministic dynamical system whose time-derivative is the expected update that the discrete process  $\theta$  would incur over one unit of time. [Theorem 1](#) guarantees that the sequence of pure-jump processes  $({}^n\theta)_{n \in \mathbb{N}}$  draws closer and closer (in probability) to the continuous process  $\Theta$ . The theorem relies on a law-of-large-numbers argument: when updates occur at high frequency and each update is small, the process behaves similarly to its expectation. Martingale methods can then be used to provide simple  $L_2$ -norm bounds on the accuracy of the approximation (see, e.g., [Darling and Norris \(2008\)](#)).

## 2.2 Reinforcers in Continuous Time

Instead of analyzing the discrete reinforcers, we focus on their fluid limit. Abusing notation, we will write  $\theta$  to denote such fluid approximation. We now introduce a number of concepts from the literature on dynamical systems that we use throughout the paper to formalize our statements.

**Definition 2.** Given a dynamical system  $\frac{d\theta}{dt}$ , the *flow* of  $\theta$  starting in  $x$  is the map

$$\begin{aligned} \theta(\cdot, x): [0, +\infty) &\rightarrow \mathcal{T} \\ t &\mapsto \theta(t, x) \end{aligned}$$

that satisfies the dynamical system equation with initial condition  $\theta(0, x) = x$ . A *trajectory* of the dynamical system is the graph of the flow,  $\{(t, \theta(t, x)): t \in [0, +\infty)\}$ . The set  $\Gamma_x = \{\theta(t, x): t \in [0, +\infty)\}$  is the *orbit* of  $\theta$  starting from  $x$ . If a sequence  $(t_n)_{n \in \mathbb{N}}$  is such that

$$\begin{cases} \lim_{n \rightarrow \infty} t_n = +\infty \\ \lim_{n \rightarrow \infty} \theta(t_n, x) = y \end{cases}$$

we say that  $y$  belongs to the *forward limit set* for the orbit  $\Gamma_x$ . A *steady state* of the dynamical system is a fixed point of its law of motion, i.e.  $\frac{d\theta}{dt}(t) = 0$ .

According to [Definition 1](#) the actions taken by a reinforcer depend directly on its current perceptions; thus, analyzing the dynamical system of  $\theta^i$  is a good proxy for the path of play. Moreover, studying the forward limit set of a reinforcer allows us to focus on estimated values instead of realized actions. Since many policies, such as  $\varepsilon$ -greedy, prescribe randomly exploring at small rates arbitrarily far in the future, even if the reinforcer settles its perceptions and identifies the action with the largest expected reward, it will keep playing sub-optimal actions (albeit with a small probability). Thus, focusing on the path of perceptions instead of the path of play provides a natural way to define stationarity in the case of learning: a reinforcer becomes stationary if its perceptions reach a steady state. We adopt this view for simplicity and clarity of exposition.

**Definition 3.** We say a reinforcer  $(\theta^i, \pi^i)$  initialized at  $x$  *converges* if  $\theta^i(t)$ 's flow started at  $x$  converges on a steady-state  $\bar{\theta}^i$ . We say the agent *converges on action*  $a_{ss}$  if the steady-state  $\bar{\theta}^i$  is such that  $a_{ss} \in \operatorname{argmax}_{a \in A_i} \bar{\theta}_a^i$ . If the  $\operatorname{argmax}$  is not a singleton, we say  $\bar{\theta}^i$  is a *pseudo-steady-state*.

Requiring a steady state to always exist can be stringent, which motivates the following definition of *learning* for this paper: agents learn action  $a_l$  if the perception of that action is always the largest in the limit.

**Definition 4.** We say the reinforcer *learns action*  $a_l$  if there exists a  $T > 0$  such that  $a_l \in \operatorname{argmax}_{a \in A} \theta_a^i(t)$  for all  $t \geq T$ . Equivalently, for all  $\bar{\theta}^i$  in the forward limit set of a given orbit,  $\bar{\theta}_{a_l}^i = \max_{a \in A} \bar{\theta}_a^i(t)$ .

It follows that if the forward limit set of  $\theta$  is a singleton, an agent who learns action  $a_l$  also converges on action  $a_l$ . Finally, notice that an agent can learn action  $a_l$  even if she doesn't play said action in each period (for example because an agent explores with positive probability in the limit).

**Separable reinforcers.** The definition of reinforcers is rather permissive, covering many procedures well known in the AI literature. For the sake of tractability, in the rest of the paper we will focus on what we call *separable* reinforcers.

**Definition 5.** A reinforcer  $(\theta^i, \pi^i)$  is said to be *separable* if there exist functions  $U$  and  $V$  such that the fluid approximation of  $\theta^i$  is of the form

$$\frac{d\theta_a^i(t)}{dt} = \alpha_a^i(\theta^i(t)) \left[ U(\theta_a^i(t), \mathbb{E}_{\pi_{-i}}[r^i(a, \pi_{-i})]) + V(\theta^i(t)) \right]. \quad (2)$$

where  $\alpha_a^i(\theta^i): \mathcal{T} \rightarrow [0,1]$ . Moreover we require that  $\alpha$ ,  $U$ , and  $V$  be a.e. Lipschitz in all components,  $U$  be Lipschitz everywhere and increasing in  $\mathbb{E}[r(a, \pi_{-i})]$  and decreasing in  $\theta_a^i$ , and  $\frac{\partial V}{\partial \theta_a^i} < -\frac{\partial U}{\partial \theta_a^i}$  almost everywhere.

Two parts make up a separable reinforcer: a component that is equal for all actions,  $V(\theta^i)$ , and a component that is action specific but Lipschitz over the entire domain  $\mathcal{T}$ ,  $U(\theta_a^i, \mathbb{E}[r^i])$ . The function  $U$ , uniform across all actions, operates on a given action's estimates. The first of the monotonicity assumptions amounts to requesting that a reinforcer's updates increase with good news. The second states that a reinforcer likes surprises: the update shrinks if the agent already holds an action in high regard.

**Example 1** (continued).  *$\varepsilon$ -greedy Q-learning is a separable reinforcer. For Q-learning,*

$$\begin{aligned} U(Q_a^i(t), \mathbb{E}[r^i]) &= \mathbb{E}_{\pi^{-i}}[r^i(a^i, a^{-i})] - Q_a^i(t) \\ V(Q^i) &= \gamma \max_a Q_a^i(t) \\ \alpha_a^i(Q^i(t)) &= \alpha^i \cdot (\pi_\varepsilon(Q^i(t)))_a \end{aligned}$$

*Thus,  $U$  is linear in its arguments and Lipschitz everywhere, increasing in rewards and decreasing in  $Q_a^i(t)$ . The common component  $V$  is constant over its Lipschitz domains and since  $\gamma < 1$  its derivative is dominated by  $U$ 's. Finally, the learning rate  $\alpha_a^i(Q^i(t))$  is the product of the common rate  $\alpha^i$  and the probability of selecting action  $a$  given  $Q^i(t)$ .*

Separability mainly requires that an algorithm “treats each action the same”: the only term that can differ across actions is the learning rate  $\alpha_a^i$ . All entries of the vector  $\theta^i$  adopt the same action-specific component, and any interaction term appears uniformly in all actions' updates. The restriction to separable reinforcers does not rule out standard algorithms such as temporal difference learners (e.g., SARSA besides Q-learning) and regret minimizers, such as the celebrated Multiplicative Weights ([Foster and Vohra \(1997\)](#) in economics, among others) and EXP3 ([Auer et al. \(2002\)](#)). Instead, it rules out mixed learning procedures that embed previous knowledge about the environment into the learning process, for example through heuristics or model-based assumptions. In all those instances, one action receives “preferential” treatment, for example by requiring that other strategies perform substantially better than a baseline before being adopted.

### 3 Q-learning in the Prisoner's Dilemma

This section analyzes how Q-learning algorithms behave in a family of Prisoner's Dilemmas to build intuition towards more general statements. Simulations show that Q-learners

|       |   |            |            |
|-------|---|------------|------------|
|       |   | Bob        |            |
|       |   | C          | D          |
| Alice | C | $2g, 2g$   | $g, 2 + g$ |
|       | D | $2 + g, g$ | $2, 2$     |

Figure 1: Payoffs of the stage game,  $1 < g < 2$ .

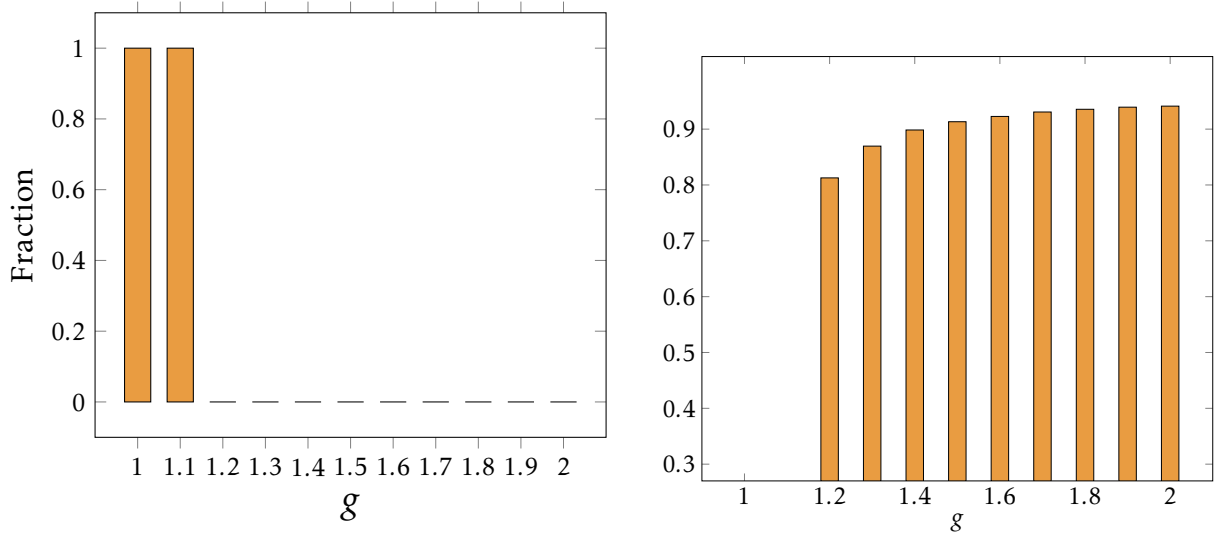
can coordinate to cooperate over a wide range of settings, but their coordination is not perfect and they randomly switch between cooperation and defection. Analyzing the fluid approximation allows us to pin down the key mechanism behind cooperation.

**Model.** We consider a family of Prisoner’s Dilemma games, as illustrated in Figure 1. Alice and Bob both start with an equal endowment of \$2 and simultaneously decide whether to invest their funds in a shared pool that grows in value by a factor of  $g$ , where  $g$  falls between 1 and 2. The accumulated wealth in the pool is then divided equally between the two, and those who didn’t invest get to keep their endowments as well. This game is a canonical model of the free-rider problem, equivalent to a Prisoner’s Dilemma. The parameter  $g$  models the attractiveness of cooperation: the larger  $g$ , the more attractive cooperation (action C) becomes. However, for all  $g \in (1, 2)$ , the dominant strategy, and the only Nash equilibrium, is to play “defect” (action D) and keep one’s change.

**Simulations.** We simulate the path of play of Alice and Bob when they adopt  $\varepsilon$ -greedy Q-learning in this free-rider problem. We run 100 simulations for each value of  $g$  on a grid between 1 and 2. In each simulation, the algorithms are initialized *optimistically*, i.e., with Q-values  $Q_C(0) < Q_D(0)$ , both larger than the maximum feasible continuation value  $\frac{2g}{1-\gamma}$ ; Alice and Bob are initialized symmetrically,  $Q_a^A(0) = Q_a^B(0)$  for both actions  $a \in \{C, D\}$ .<sup>7</sup> The algorithms interact over the course of  $10^5$  periods. Both Alice and Bob adopt a Q-learning algorithm with  $\varepsilon = 0.1$ ,  $\alpha = 0.05$  and  $\gamma = 0.9$ . We will frequently refer to this parametrization as the *standard parametrization*, where both agents experiment somewhat frequently (in 5% of the interactions) but the perceptions are persistent.

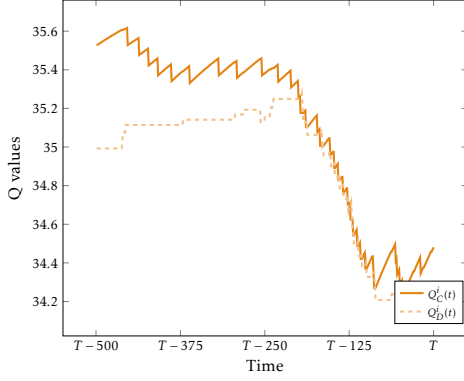
Figure 2a shows the results of these numerical experiments. The algorithmic agents learn to play the dominant strategy equilibrium  $\{D, D\}$  only for low values of the parameter  $g$ . Instead, when  $g$  is large, the agents cooperate, albeit imperfectly. Both cooperation and defection appear in unpredictable but recurrent cycles, as shown in Figures 3a

<sup>7</sup>Optimistic initialization is common in the reinforcement learning literature to stimulate experimentation in the early stages of the learning process. Later in this section we present results with alternative initialization (including randomized), and we show how our theory helps predict the effect of initialization on outcomes.

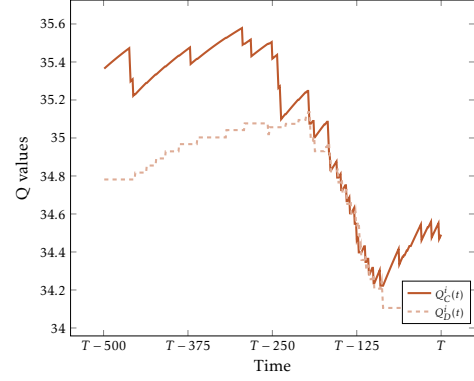


(a) Each bar represents the percentage of simulations for each  $g$  where the agents learn to play the Nash Equilibrium  $\{D, D\}$  (i.e., agents play the pair  $\{D, D\}$  in over 50% of the last  $2 \cdot 10^4$  iterations). (b) Each bar represents the ratio of periods  $k \in [8 \cdot 10^4, 10^5]$  where cooperation is a player's preferred action ( $Q_C(k) \geq Q_D(k)$ ) over the total number of periods in the same interval ( $2 \cdot 10^4$ ), for a sample experiment with parameter  $g$ .

Figure 2: Results of simulations



(a) Evolution of Alice's Q-values.



(b) Evolution of Bob's Q-values.

Figure 3: The two graphs depict the last 500 iterations of a sample simulation with standard parametrization and  $g = 1.8$ .

and 3b: the value of collaboration is generally above that of defection, but after some time it drops below  $Q_D(k)$ , so that agents switch to playing  $D$ . The value of defection then decreases almost immediately, and players revert to cooperation. These simulations highlight a few puzzles. First, cooperation arises only for large values of  $g$ , even though defection is always a dominant strategy. Second, cooperation seems to consist of cycles, but one cannot impute these to “retaliatory” strategies, since Q-learning does not carry memory of past play. Figure 2b shows that in the long run Alice and Bob play cooperation for a large fraction of the time.

### 3.1 Theoretical Results

We first explicitly derive the continuous-time approximation of  $\varepsilon$ -greedy Q-learning. In every iteration, Alice and Bob play (with high probability) the action corresponding to their largest perception. Both learn something about its value and then update their perception. In the parlance of Definition 1, Alice and Bob evolve according to a function  $D^i$  that is discontinuous along the surface  $Q_D^i(k) = Q_C^i(k)$ . To address the discontinuity of the discrete-time system, we first apply Theorem 1 to  $Q(k)$  over the largest open sets such that  $D = (D^A, D^B)$  is everywhere Lipschitz. We call these sets *maximal continuity domains*: in the case of  $\varepsilon$ -greedy Q-learning these are sets  $\omega_{a,b}$  of 4-dimensional vectors  $Q$  such that Alice's greedy action is  $a$  and Bob's greedy action is  $b$ ; let  $\bar{\omega}_{a,b}$  be their closure.

In  $\omega_{C,C}$  the greedy action for both players is  $C$ , so that in every period Alice cooperates with probability<sup>8</sup>  $1 - \frac{\varepsilon}{2}$  and defects with probability  $\frac{\varepsilon}{2}$ . Hence, with probability  $(1 - \frac{\varepsilon}{2})^2$

<sup>8</sup>That is,  $C$  is selected with probability  $1 - \varepsilon$  if the randomization device instructed the agent to be

she collects reward  $2g$  — similarly for other profiles. The fluid limit in  $\omega_{C,C}$  solves

$$\begin{cases} \frac{d\mathbf{Q}_C^A(t)}{dt} = \alpha \left(1 - \frac{\varepsilon}{2}\right) \left[ \left(1 - \frac{\varepsilon}{2}\right) 2g + \frac{\varepsilon}{2}g + (\gamma - 1)\mathbf{Q}_C^A(t) \right] \\ \frac{d\mathbf{Q}_D^A(t)}{dt} = \alpha \frac{\varepsilon}{2} \left[ \left(1 - \frac{\varepsilon}{2}\right) (2 + g) + 2\frac{\varepsilon}{2} + \gamma\mathbf{Q}_C^A(t) - \mathbf{Q}_D^A(t) \right] \\ \frac{d\mathbf{Q}_C^B(t)}{dt} = \alpha \left(1 - \frac{\varepsilon}{2}\right) \left[ \left(1 - \frac{\varepsilon}{2}\right) 2g + \frac{\varepsilon}{2}g + (\gamma - 1)\mathbf{Q}_C^B(t) \right] \\ \frac{d\mathbf{Q}_D^B(t)}{dt} = \alpha \frac{\varepsilon}{2} \left[ \left(1 - \frac{\varepsilon}{2}\right) (2 + g) + 2\frac{\varepsilon}{2} + \gamma\mathbf{Q}_C^B(t) - \mathbf{Q}_D^B(t) \right] \end{cases} \quad (3)$$

where we denote in bold the fluid approximation of the algorithms of Alice and Bob. Similar linear systems appear in all continuity domains. We can write the dynamical system in matrix form as

$$\dot{\mathbf{Q}}(t) = \begin{cases} A_{C,C}\mathbf{Q}(t) + b_{C,C} & \text{for } \mathbf{Q}(t) \in \omega_{C,C} \\ A_{C,D}\mathbf{Q}(t) + b_{C,D} & \text{for } \mathbf{Q}(t) \in \omega_{C,D} \\ A_{D,C}\mathbf{Q}(t) + b_{D,C} & \text{for } \mathbf{Q}(t) \in \omega_{D,C} \\ A_{D,D}\mathbf{Q}(t) + b_{D,D} & \text{for } \mathbf{Q}(t) \in \omega_{D,D} \end{cases} \quad (4)$$

The theory of differential inclusions (Filippov (1988)) guarantees that there exists a (forward in time) solution to this dynamical system with discontinuous right-hand side.

The evolution of this 4-dimensional piecewise-linear system is still complex, because the system generally exhibits chaotic behavior (see Appendix B for a formal analysis of chaos). Moreover, while the theory of differential inclusions suffices to establish existence of a solution, it does not establish uniqueness; in fact, the system in Equation (4) admits infinitely many solutions. To make progress we need to select one of these solutions, and because Alice and Bob are *ex-ante* symmetric, a natural assumption to impose is that of symmetry. Specifically, we restrict attention to the subspace  $\{Q \in \mathbb{R}^4 | Q_a^A = Q_a^B \text{ for } a = C, D\} \cong \mathbb{R}^2$ , where the system is bound to remain if the initial condition is symmetric. That is, we make the following assumption for the rest of Section 3:<sup>9</sup>

**Assumption S.** Let  $\mathbf{Q}_a^A(0) = \mathbf{Q}_a^B(0)$  for  $a = C, D$ .

In words, this assumption requires that Alice and Bob’s algorithms be initialized symmetrically. In Appendix B we analyze in detail the dynamics of the 4-dimensional system and discuss which theoretical insights can be proved without Assumption S; moreover,

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greedy and with probability  $\frac{\varepsilon}{2}$  if the agent plays a random action.

<sup>9</sup>We will be explicit on which results do not rely on it.

for every result that we prove in the rest of this section, [Appendix B](#) provides either a theoretical or numerical counterpart. We argue there that, while this assumption is not without loss, the insights we derive under it shed light on the most surprising behaviors that we observe in simulations.

Under [Assumption S](#), we prove the following proposition, which characterizes the limiting behavior of the continuous-time approximation.

**Proposition 1.** *Let  $\underline{\varepsilon}(g) = 1 - \sqrt{\frac{2-g}{g}}$ . The forward limit set of  $\mathbf{Q}$  is a singleton for any initial condition. If  $\varepsilon \geq \underline{\varepsilon}(g)$ , all initial conditions lead to the unique steady state*

$$q_D^{eq} = \left( \frac{2g + \varepsilon g}{2} + \frac{\gamma(4 + \varepsilon g)}{2(1 - \gamma)}, \frac{4 + \varepsilon g}{2(1 - \gamma)} \right)$$

which lies in  $\omega_{D,D}$ .

If  $\varepsilon < \underline{\varepsilon}(g)$ , initial conditions may lie in one of two regions of attractions,  $R_D$  and  $R_C$ . All initial conditions in  $R_D$  lead to the steady state  $q_D^{eq}$ . All initial conditions in  $R_C$  instead lead to the pseudo-steady state (see [Definition 3](#))

$$q_C^{eq} = (y, y) \text{ where } y = \frac{1 + g + \sqrt{(g-1)(g-1-\varepsilon g + \frac{\varepsilon^2 g}{2})}}{(1-\gamma)},$$

which lies in  $\overline{\omega}_{D,D} \cap \overline{\omega}_{C,C}$ . Moreover,  $R_C$  expands as  $g$  increases and shrinks as  $\varepsilon$  increases.

In the symmetric subspace, the limiting behavior of the system can be twofold. When  $\varepsilon$  is greater than the critical level  $\underline{\varepsilon}(g)$ , there is a unique steady state  $q_D^{eq}$  that algorithms always converge to, where both players consider defection as their preferred action. Instead, when  $\varepsilon$  is below the critical level, there exists an additional steady state on the boundary that separates the two regions  $\omega_{D,D}$  and  $\omega_{C,C}$  — a pseudo-steady-state according to [Definition 3](#), where Alice and Bob's perceptions of cooperation and defection coincide.

In this case the perceptions lie between the long-run value of mutual defection,  $\frac{r(D,D)}{1-\gamma}$ , and the long-run value of mutual cooperation,  $\frac{r(C,C)}{1-\gamma}$ . Alice and Bob are indifferent between cooperation and defection at  $q_C^{eq}$ , and they play cooperation (resp., defection) for a fraction  $\tau_c$  (resp.,  $1 - \tau_c$ ) of the time.

**Corollary 1.** *In the pseudo-steady-state  $q_C^{eq} = (y, y)$  agents spend  $\tau_c$  fraction of their time cooperating, where*

$$\tau_c = \frac{\frac{\varepsilon^2 g}{2} + \varepsilon - 2 - y(\gamma - 1)(1 - \varepsilon)}{2(\varepsilon - 1)(1 + g + (\gamma - 1)y)} \in \left[ \frac{1}{2}, 1 \right].$$

The pseudo-steady-state  $q_C^{eq}$  corresponds to the imperfect cooperation we observed in the experiments. In particular, the analytic expression for the time spent cooperating closely approximates its discrete-time experimental counterpart, as shown in [Figure 8b](#).<sup>10</sup>

### 3.2 Sketch of the Proofs

Under [Assumption S](#), learning evolves according to the following piecewise-linear dynamical system:

$$\dot{\mathbf{Q}}(t) = \begin{cases} A_{C,C}\mathbf{Q}(t) + b_{C,C} & \text{for } \mathbf{Q}(t) \in \omega_{C,C} \\ A_{D,D}\mathbf{Q}(t) + b_{D,D} & \text{for } \mathbf{Q}(t) \in \omega_{D,D} \end{cases}$$

The flows within each  $\omega_{a,a}$  characterize the evolution of the  $\mathbf{Q}$ -vector. However, the system is discontinuous: we would like to preserve continuity of any given flow along the boundary  $\overline{\omega}_{C,C} \cap \overline{\omega}_{D,D}$ . [Proposition 2](#) below guarantees that we can suitably extend the flows on the boundary, similarly to continuous pasting techniques, such that the flow defined by the fluid limit is globally well-defined forward in time.

**Proposition 2.** *Let  $F_a$  be the field defined as above over  $\omega_{a,a}$  for all  $a \in \{C, D\}$ . There exists a global solution in the sense of [Filippov \(1988\)](#) to the differential inclusion*

$$\begin{aligned} \frac{d\mathbf{Q}_t}{dt} &= F_a(\mathbf{Q}_t) && \text{over } \omega_{a,a} \text{ for } a = C, D \\ \frac{d\mathbf{Q}_t}{dt} &\in \text{co}\{F_a(\mathbf{Q}_t) \mid \forall a = C, D\} && \text{when } \mathbf{Q}_t \in \overline{\omega}_{C,C} \cap \overline{\omega}_{D,D} \end{aligned}$$

where  $\text{co}\{\cdot\}$  denotes the convex hull of a set, and  $\overline{\omega}$  is the closure of  $\omega$ .

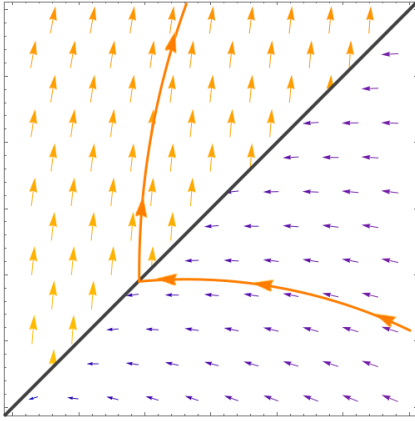
Both vector fields  $F_C$  and  $F_D$  are well-defined also on the boundary between  $\omega_{C,C}$  and  $\omega_{D,D}$ . This boundary is called a *switching surface*, because the laws of motion must switch from one field to the other. Adopting the convention of Filippov, we can define a vector field on the switching surface such that all flows extend to a global solution and such that certain continuous-pasting conditions are satisfied.<sup>11</sup>

Let us call  $\hat{\mathbf{N}}$  the unit normal vector to the switching surface  $\overline{\omega}_{C,C} \cap \overline{\omega}_{D,D}$ , so that  $\hat{\mathbf{N}} \cdot F_a$  is the component of  $F_a$  orthogonal to the surface. We divide the switching surface in three regions, according to the signs of the normal components of the two vector fields

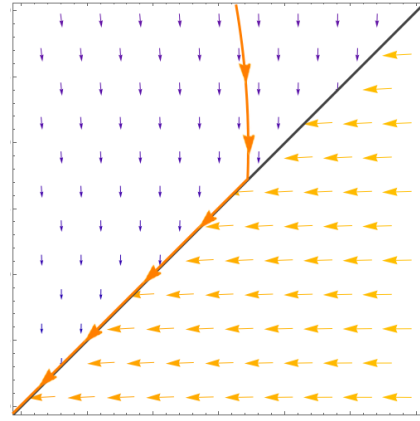
<sup>10</sup>In [Appendix B](#) we observe that the general asymmetric 4-D system gravitates around a similar equilibrium, where  $Q_C = Q_D$ . We interpret it as additional support for the restriction to a symmetric subspace.

<sup>11</sup>Uniqueness in general is not guaranteed: the behavior on the switching surface can lead to multiplicities in the behavior of the system. The law of motion on the switching surface needs only to belong to the convex hull of the laws of motion on either side; by constraining the motion to satisfy certain continuous-pasting conditions we obtain well-defined orbits.

$F_C$  and  $F_D$ . A *crossing region* occurs when both normal fields to the boundary are of the same sign: either  $F_C$  or  $F_D$  can be used to define the law of motion on the surface, since any orbit leaves the switching boundary immediately. A *repulsive region* occurs where both normals face away from the boundary, which will then never be reached; unless initialized here, an orbit will never intersect the repulsive region, thus we do not need to define a field here. Finally, a *sliding region* occurs when both normal components of the two vector fields point towards the boundary. Every flow hitting the switching surface in the sliding region must continue on the switching surface, *sliding along the boundary*. The sliding vector field is defined as a convex combination of the two vector fields  $F_C$  and  $F_D$  with parameter  $\tau$  such that the normal component to the switching surface vanishes, i.e.  $\tau(\hat{\mathbf{N}} \cdot F_C) + (1 - \tau)(\hat{\mathbf{N}} \cdot F_D) = \vec{0}$ . The vector field  $\tau F_C + (1 - \tau)F_D$  is the unique vector field in the convex hull of  $F_C$  and  $F_D$  whose flow is confined to the switching surface and that satisfies the differential inclusion requirements. Figure 4 plots the vector fields around the boundary and shows examples of sliding and crossing boundaries with a sample orbit.



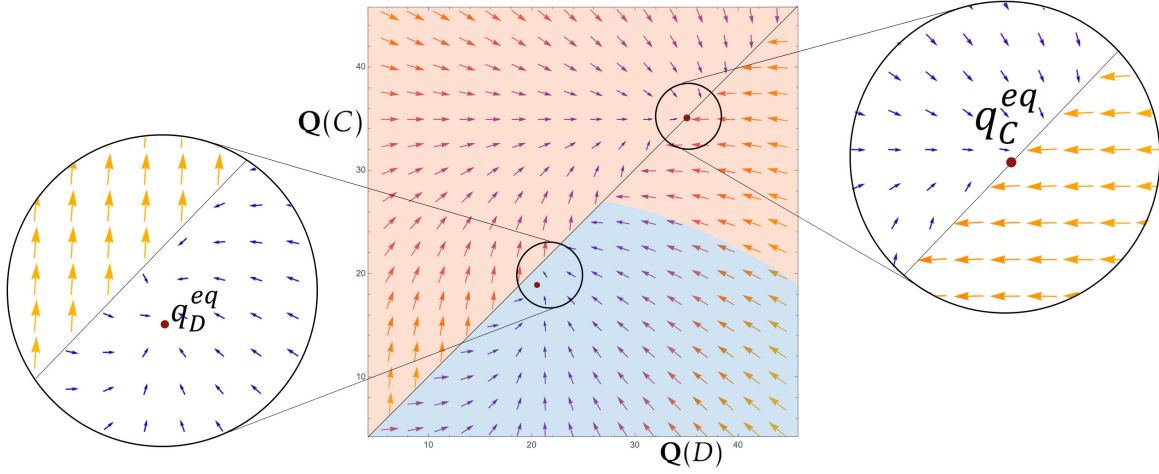
(a) Orbit crossing the boundary.



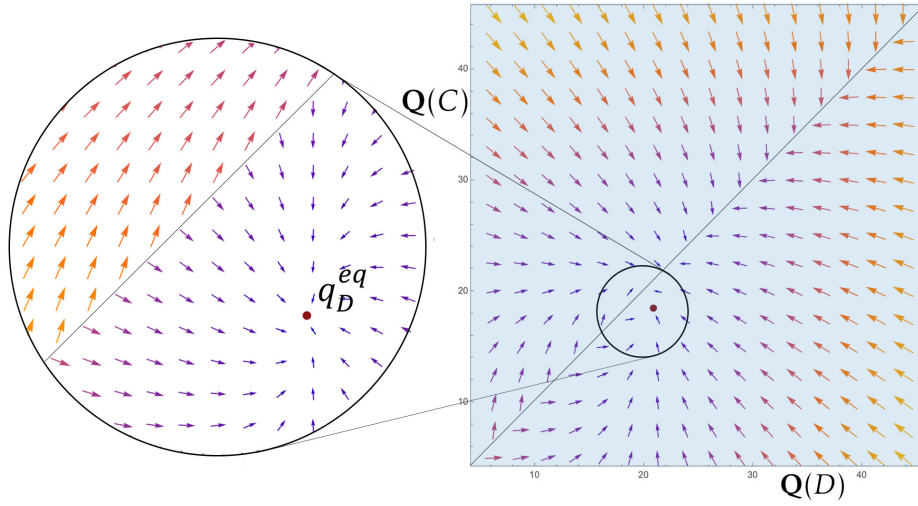
(b) Orbit sliding along the boundary.

Figure 4: Depiction of two discontinuous flows around a switching surface, in the crossing and sliding case.

One can then look for stationary points by finding the solutions to  $F_C = 0$ ,  $F_D = 0$ , and  $\tau F_C + (1 - \tau)F_D = 0$ . We show that  $F_C = 0$  never has a solution in  $\omega_{C,C}$ ,  $F_D = 0$  always have a unique one in  $\omega_{D,D}$ , and that  $\tau F_C + (1 - \tau)F_D = 0$  has a solution only when  $\varepsilon \geq \underline{\varepsilon}(g)$ . In Figure 5 we plot the vector fields that define  $\mathbf{Q}$  for two different values of  $g$ : notice two stationary points when  $g$  is large — the pseudo-steady-state on the boundary disappears for low values of  $g$ . Figure 6 similarly depicts the attraction regions when  $g = 1.8$ , and shows how simulations stack against analytical predictions. For the right panel, we record the results of 50 training runs for each initialization point on the grid  $Q_0(D), Q_0(C) = 15, 16, \dots, 40$ , and we check whether the algorithms cooperate over 50% of



(a) Phase space with  $g = 1.8$ .



(b) Phase space with  $g = 1.1$ .

Figure 5: Stationary points are marked with a red dot. The domain of attraction of the cooperative outcome is orange-shaded, and the one for the non-cooperative outcome is blue-shaded.

the time.<sup>12</sup> When the initialization falls in the attraction region of cooperation,  $R_C$ , algorithms cooperate in all training runs. When initialization falls instead in  $R_D$ , cooperation appears in only a fraction of training runs. For details on the analytical characterization of [Proposition 1](#) we refer to [Appendix A](#).

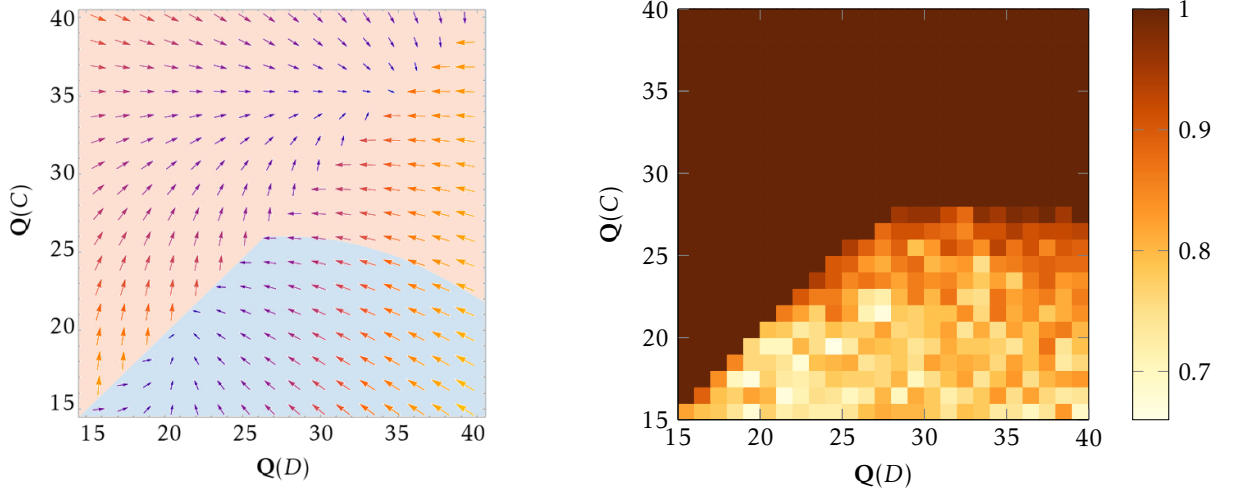


Figure 6: The left panel shows the attraction region for the algorithms with standard parametrization and an attractiveness of cooperation  $g = 1.8$ . The region  $R_C$  is pink, while the region  $R_D$  is blue. The right panel shows the empirical probability that two algorithms initialized in  $(Q_0(D), Q_0(C))$  with standard parametrization support cooperation through spontaneous coupling.

### 3.3 Interpretation

According to [Proposition 1](#), the *cooperative steady-state*,  $q_C^{eq}$ , exists only when the exploration rate is small enough. [Figure 7](#) shows the threshold level of exploration  $\underline{\varepsilon}$  that guarantees a cooperative steady-state and how it varies with the value of cooperation. Intuitively, as  $g$  increases, the relative benefit of defecting decreases (and vanishes for  $g = 2$ ), so more and more exploration is needed to realize that  $D$  is a dominant action. For example, if  $g = 1.8$  the exploration rate required to guarantee convergence on  $\{D, D\}$  is about 70%, which is considerably larger than the standard employed in practice.<sup>13</sup>

<sup>12</sup>Consistently with the simulations of [Figures 2a](#) and [2b](#), Alice and Bob’s Q-values are initialized symmetrically. However, the discrete system’s randomness, caused by exploration, breaks symmetry almost immediately.

<sup>13</sup>The literature on Q-learning in games usually employs  $\varepsilon = 0.1$  or smaller, either fixed or decreasing over time. See, e.g., [Gomes and Kowalczyk \(2009\)](#).

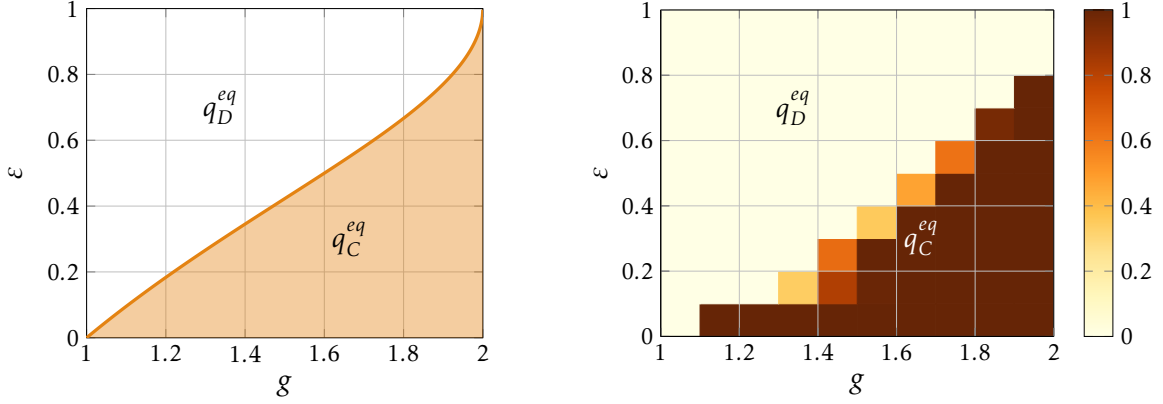


Figure 7: The left panel shows the maximum exploration rate  $\underline{\varepsilon}(g) = 1 - \sqrt{\frac{2-g}{g}}$  to support the cooperative equilibrium as a function of the attractiveness of cooperation. The right panel shows the fraction of training runs with standard parametrization that support cooperation through spontaneous coupling for a grid of parameters  $\varepsilon = 0, 0.1, 0.2, \dots, 1$  and  $g = 1, 1.1, 1.2, \dots, 2$ .

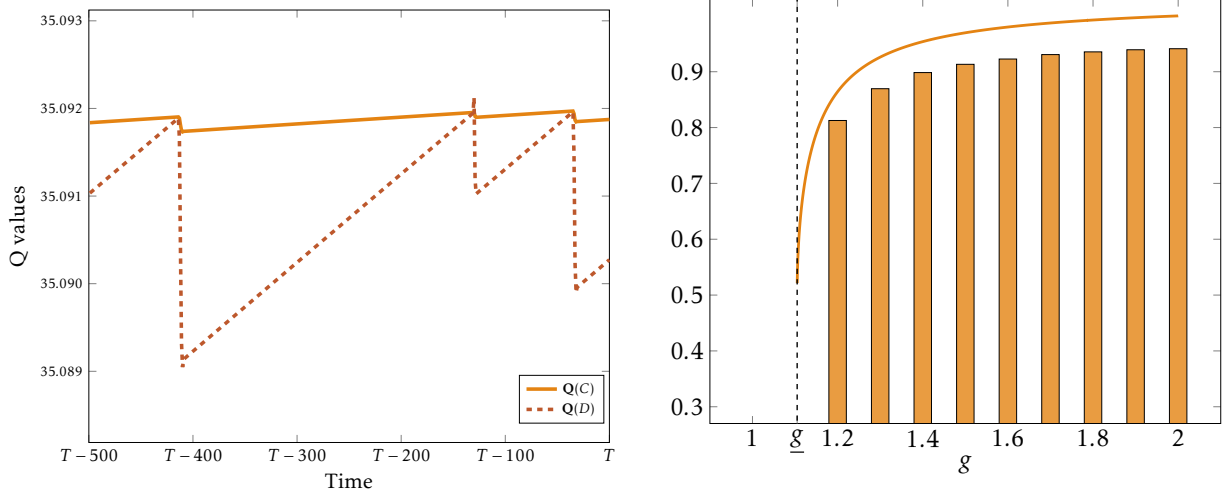
How does Q-learning sustain long-run cooperation, and why does the exploration rate matter? We answer these questions by analyzing the cooperative steady state, and its representations in Figure 8a together with Figures 3a and 3b.<sup>14</sup> Suppose C is the preferred action of both players: Alice and Bob cooperate, but with probability  $\frac{\varepsilon}{2}$  one defects and is “surprised” by the benefit of defecting unilaterally. The deviating agent enjoys a surprise of size  $\Delta_{CD}^i = 2 + g - 2g = 2 - g$ . Through repeated experiments, such surprises increase the perception  $Q_D^i$  to a level above  $Q_C^i$ . Suppose this happens first to Alice. When Alice defects, Bob experiences a sharply negative surprise of size  $\Delta_{CD}^- = g - 2g = -g$ . Therefore, soon after Alice begins defection, Bob defects too — cooperating makes Bob considerably worse off when Alice defects, and his perception  $Q_C^B$  quickly falls below  $Q_D^B$ . But now, mutual defection decreases the perception of defection  $Q_D^i$  for Alice and Bob, with a joint surprise roughly of size  $2 - 2g < 0$ . Naturally, when experimenting with cooperation they would experience an even worse surprise for their perception  $Q_C^i$ , of size  $g - 2g = -g$ . However, experiments are so infrequent that  $Q_C^i$  changes very slowly. Consequently, after a brief phase of joint defection,  $Q_D^i$  will fall below  $Q_C^i$  again, restarting the cycle.

Effectively, when the exploration rate is low, the AI agents play a symmetric profile of actions too often. This is the type of statistical linkage between algorithm’s estimates that we call *spontaneous coupling* — the perceptions of independent Q-learners tend to evolve

<sup>14</sup>Notice that, when simulated with small but discrete time steps, the dynamical system closely mimics the path of play of the discrete Q-learning.

symmetrically.<sup>15</sup> When algorithms start defecting, cooperation experiments become rare, which makes the perception  $Q_C^i$  of cooperation's value persistent. Alice's estimate  $Q_C^A$  remains close to the long-run value of mutual cooperation. Instead, her perception  $Q_D^A$  drops dramatically once both agents begin defecting. Alice never realizes the downside of cooperating when Bob defects because both revert to cooperation simultaneously.

The attractiveness of cooperation is directly connected to the time spent cooperating. Large values of  $g$  imply a small surprise  $\Delta_{CD}^i$ , which delays the start of a defection episode. Once an agent defects, the surprise  $\Delta_{CD}^i$  is larger the larger the parameter  $g$ , which accelerates the response from the other agent. Periods of mutual defection shorten, because the joint surprise gets larger for larger values of  $g$ . This is reflected in the statistic reported by Figure 8b.



(a) Cycles of play around the cooperative equilibrium in the discretized ODE system.

(b) Proportion of time spent playing the cooperative outcome in  $q_C^{eq}$ , analytically and in the simulations.  $\underline{g}$  is the lowest attractiveness of cooperation such that spontaneous coupling arises when  $\varepsilon = 0.1$ , defined as  $\underline{g} = (\bar{\varepsilon})^{-1}(0.1)$ .

Figure 8

The pseudo-steady-state on the boundary is the continuous-time counterpart of these cycles in the discrete system. In other words, the discrete stochastic cycles converge to a single point in the continuous-time limit. In the pseudo-steady-state, agents are “indifferent” between cooperation and defection. Both spend some “local time” playing either action: we interpret the weights  $\tau_c$  and  $1 - \tau_c$  assigned to the fields on the boundary as

<sup>15</sup>Naturally, the continuous-time approximation together with Assumption S exacerbate the symmetry of play. However, the intuition we delineated so far does not rely on the Q-values being identical across players. In fact, in Appendix B we find that similar behavior arises when we dispense of Assumption S.

the fraction of time dedicated to cooperation and defection, respectively.<sup>16</sup> The local time captures the characteristics of discrete cycles. It is such that the infinitesimal incentives around the stationary point are balanced.

**Robustness.** We imposed [Assumption S](#) at the beginning of the formal results of this section. While symmetry is necessary to gain analytical tractability, spontaneous coupling appears even without this assumption. To see this, consider the discrete stochastic system. Even when initialized symmetrically, symmetry breaks as early as the first period of random experimentation. Nonetheless, in [Figure 8b](#) the expression  $\tau_c$ , obtained through the symmetric approximation, appears to fit the discrete, stochastic data fairly well. [Figure 6](#) and [Figure 7](#) show a similar pattern when varying the symmetric initializations. In [Appendix B](#) we study in greater detail the dynamical system that arises without symmetric initialization. We show that without [Assumption S](#) tractability is lost because (i) a unique solution to the differential inclusion can no longer be guaranteed and (ii) the resulting system is chaotic ([Holden \(2014\)](#)). Nevertheless, we argue that spontaneous coupling still explains the dynamics of cooperation; in particular, we show that cooperation occurs because players are coupled in their *switching* from cooperation to defection (and viceversa), and not because their perceptions are equal. At the same time, symmetric initialization is a restrictive assumption, in that the range of initial values leading to cooperation may be smaller when the Q-learners are initialized at different points.

A common feature of simulation studies about algorithmic collusion is the observation that coordination tends to break down as the number of agents involved increases (e.g., [Waltman and Kaymak \(2008\)](#)). Our framework allows us to seamlessly study an extension of the Prisoner’s Dilemma of this section to a setting with arbitrarily many players, which we pursue in [Appendix C](#). Under the equivalent of [Assumption S](#) we prove that the number of players can indeed affect the emergence of coordination for  $N \geq 2$  Q-learners. In particular, we find that: (i) when the exploration rate is neither very high nor very low, the number of players  $N$  is the determining factor of whether a coordinating pseudo-steady-state exist;<sup>17</sup> (ii) even when the pseudo-steady-state exists, the local time spent cooperating decreases with  $N$ . The intuition is that as the number of players increases, the likelihood that at least one of them will defect while “cooperate” is the preferred action increases as well. This, in turn, decreases the payoff perceived by jointly cooperating players, eventually making cooperation impossible.

<sup>16</sup>See [di Bernardo et al. \(2008\)](#) for a formal treatment of this intuition.

<sup>17</sup>If  $N$  is higher than a threshold that depends on both the exploration rate and the attractiveness of cooperation the pseudo-steady-state does not exist.

**Spontaneous Coupling as Collusion?** The Artificial Intelligence algorithms we consider sustain supracompetitive outcomes without explicit reward-punishment schemes and lack basic tools, like monitoring, to coordinate agreements. In our example, spontaneous coupling leads agents to joint cooperation, without communication or explicit hard-coding of cooperative incentives in the agents’ strategies. Alice and Bob stick to long periods of cooperation not under the threat of defection, but because past experience has reinforced their perception that cooperation dominates defection. By creating statistical linkages among different firms’ perceptions, spontaneous coupling provides implicit self-confirming beliefs. Alongside the mechanism in [Hansen, Misra and Pai \(2021\)](#), this is a collusive scheme that is exclusive to AI algorithms. Unlike institutional definitions of collusion, coordination via spontaneous coupling does not require explicit intent (e.g., Alice may not even be aware that there is a Bob to coordinate with).

Finally, notice that the paths of play of spontaneously coupled learning algorithms are observationally equivalent to the paths of play of settings with imperfect monitoring or non-stationary stage-games: long streaks of supra-competitive outcomes followed by short and intense price wars such as in [Green and Porter \(1984\)](#). As in [Green and Porter \(1984\)](#), price wars enforce cooperation, but unlike that setting they are not triggered by exogenous market fluctuations. Instead deviations occur endogenously as agents realize their unilateral benefit from abandoning the implicit cartel. Moreover, we observe that spontaneous coupling (and related cycles) emerge *entirely independently* of the discount factor. The cycles we observe depend on the agents’ bounded rationality instead of their foresight. Using perceptions to guide their play, they “project” into the future payoffs they earned in the past, making incorrect ex-ante inference about the returns to cooperation and defection (which turns out to be confirmed ex-post).

## 4 General Reinforcers

The mechanics of spontaneous coupling, identified in [Section 3](#), appear tied to the policy of the reinforcer. We now show that the driving force of inadvertent coordination lies in the algorithm’s uneven learning rates across different actions, of which the exploration rate is just one determinant. Learning at different rates about different actions facilitates coordination for general reinforcers. We establish a sufficient condition on the parameters of the algorithm which avoids coupling: learning rates must be uniform across actions.

## 4.1 Reinforcers with Uneven Learning Rates

The analysis in the Prisoner's Dilemma of [Section 3](#) exposes a mechanism which sustains dominated outcomes between algorithms. When Alice's preferred action is cooperation, she learns about the payoffs of cooperating at rate  $1 - \frac{\varepsilon}{2}$ , while much more slowly, at rate  $\frac{\varepsilon}{2}$ , about defection. Essentially, exploration regulates how quickly Alice learns about the value of deviations. When algorithms are coupled, low rates of exploration impair the ability of the algorithm to correctly estimate the value of deviations.

This phenomenon can be formalized without referring to a specific policy. In fact, a given policy only affects the learning rates  $\alpha_a$  of different actions. For example, in the case of  $\varepsilon$ -greedy Q-learning the differential equation within the maximal continuity domain  $\omega_{D,D}$  reads as:

$$\begin{cases} \frac{dQ_C^A}{dt}(t) = \alpha \frac{\varepsilon}{2} \left[ \left(1 - \frac{\varepsilon}{2}\right)g + \frac{\varepsilon}{2}2g + (\gamma - 1)Q_C^A(t) \right] \\ \frac{dQ_D^A}{dt}(t) = \alpha \left(1 - \frac{\varepsilon}{2}\right) \left[ \left(1 - \frac{\varepsilon}{2}\right)2 + \frac{\varepsilon}{2}(2 + g) + \gamma Q_C^A(t) - Q_D^A(t) \right] \end{cases}$$

In the language of separable reinforcers, the decision rule of player  $i$  affects only  $\alpha_a^i(\theta^i)$ . In the fluid approximation the only role of the policy is to determine the relative learning rates, since in [Definition 5](#) expectation is taken only with respect to distribution over profiles generated by the opponents' policies. Since the absolute magnitude of the learning rates does not matter, we focus on the relative rates.

**Definition 6.** The relative learning rate of action  $a^i$  is the ratio

$$RLR(a^i) = \frac{\alpha_{a^i}}{\sum_{a \in A_i} \alpha_a}.$$

[Theorem 2](#) below shows how differences in RLR across actions generally give rise to spontaneous coupling: the coupling appears in a Prisoner's Dilemma for any pair of agents using any (separable) reinforcer. For simplicity, we focus on reinforcers with maximal continuity domains equal to  $\omega_{C,C}$  and  $\omega_{D,D}$ , and constant learning rates  $\alpha_C, \alpha_D$ .

**Theorem 2.** *Let each agent learn through the same greedy reinforcer in a general Prisoner's Dilemma, and let [Assumption S](#) be satisfied. There exist an open set  $A \subset \mathbb{R}_+^4$  such that for all parameters  $\{\alpha_j(\omega_{k,k})\}_{j,k=C,D} \in A$  there exists a pseudo-steady-state on the boundary  $\overline{\omega}_{C,C} \cap \overline{\omega}_{D,D}$ .*

The proof is constructive: we show that there exists a symmetric tuple of  $\alpha$ 's such that the sliding vector field on the indifference boundary is null. We apply homotopical argu-

ments to show existence, and then perturb the problem to obtain the result by continuity. Finally, we notice that the proof applies also when learning rates depend on perceptions  $\theta$ , because the argument relies only on studying a neighborhood of the boundary.

[Theorem 2](#) restricts the occurrence of spontaneous coupling only to algorithms with discontinuous action-selection policies such as  $\varepsilon$ -greedy. This is because the proof rests on the continuous-time limit: in the approximation, smooth policies (e.g., Boltzmann exploration) preclude the formation of a pseudo-steady-state because they provide a continuous interpolation between vector fields at the switching manifold. However, this distinction is less salient in discrete-time implementations. For sufficiently low temperatures, the Boltzmann policy approximates the  $\varepsilon$ -greedy reinforcer, with the difference being confined to a small  $\delta$ -neighborhood of the switching boundary. Because discrete updates typically cause estimates to jump across this switching "tube" rather than reside within it, the resulting discrete cycles mirror those produced by discontinuous policies. A formal characterization of the "sharpness" of transitions required to trigger spontaneous coupling remains an open problem for finite-time analysis.

[Theorem 2](#) shows that spontaneous coupling arises in a Prisoner's Dilemma, but pseudo-steady-states may exist in other games too. In fact, formally, a solution to the differential inclusions on the switching manifold is a convex combination of vector fields corresponding to specific action profiles. The resulting steady states can be interpreted as coarse-correlated equilibria (CCEs): distributions over action profiles where no player has an ex-ante profitable deviation (see [Nisan et al., 2007](#)). Generally the set of CCEs can be fairly large, and therefore in general games we should expect pseudo-steady-states (i.e., spontaneous coupling) to exist. However, the Prisoner's Dilemma admits a *unique CCE*, which is mutual defection. Therefore, that spontaneous coupling sustains cooperation in this context is a distinct algorithmic anomaly rather than a standard correlation result, which arises because coordination is not underpinned by standard rationality.

It is the unique payoff structure of the Prisoner's Dilemma (one Nash equilibrium which is Pareto-dominated by joint coordination) that drives the occurrence of spontaneous coupling of separable reinforcers *beyond standard rationality predictions*. To see why, consider any other symmetric  $2 \times 2$  game. In "Deadlock" games (unique Nash equilibrium such that it is payoff-dominant), the Nash equilibrium acts as a global attractor that precludes cycles by making perceptions "sticky" to the dominant strategy. In "Stag Hunt" coordination games (two pure symmetric Nash equilibria and a mixed one), a pseudo-steady-state exists and corresponds to the unstable mixed-strategy Nash equilibrium. Finally, in "Hawk-Dove" games (two pure asymmetric Nash equilibria and a

mixed one), many pseudo-steady-states may exist, each corresponding to a CCE.<sup>18</sup> While in these instances the learning process appears to select one outcome out of the set CCEs, in the Prisoner’s Dilemma spontaneous coupling leads *outside* the set of CCEs.

**Market splitting.** We derived [Theorem 2](#) in the simple context of a Prisoner’s Dilemma, which features a limited action space. We already discussed similar spontaneous coupling in a multi-player extension of the Prisoner’s Dilemma (see [Appendix C](#)), but unfortunately larger games tend to yield underdetermined dynamical systems, even under our symmetry assumptions. To strengthen our claim that spontaneous coupling can sustain coordination in disparate settings,<sup>19</sup> we also study a model of market splitting. [Appendix D](#) provides all the details, while here we sketch the model and results. We cast the practice of market splitting in the context of keyword advertising: there are two advertisers who repeatedly bid for ad slots on three different keywords (one auction per keyword, three auctions run at every time); advertisers have heterogeneous valuations for each ad slot, in the sense that they have different clickthrough rates for displaying an ad related to a keyword. At each time they must decide for which subset of keywords to bid (i.e., they have eight actions at their disposal, one for each subset in the power set of the keywords), and we choose parameters so that in our model it is a dominant strategy to bid on all keywords. However, simulations show that  $\epsilon$ -greedy Q-learning algorithms learn to “split the market”, which in this case is realized by each advertiser bidding only for the keyword most valuable to them, thereby decreasing competition on each other’s preferred keyword and causing lower revenues for the ad platform. Inspecting the path of play, we observe the familiar pattern of cycles: advertisers generally coordinate on their preferred keyword, but at regular intervals coordination breaks down, and it is followed by short bouts of competition where both advertisers bid on all keywords, only for coordination to be restored. [Figure 16](#) in [Appendix D](#) represents this familiar dynamic in the space of Q-values (similarly to [Figures 3a](#) and [3b](#), although the action space is now much larger).

## 4.2 Reinforcers with Uniform Learning Rates

We argued that the emergence of coordination driven by spontaneous coupling depends on some intrinsic features of learning algorithms, their relative learning rates. In this

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<sup>18</sup>The set of CCEs for Hawk-Dove is a polytope with non-empty interior, i.e., there are uncountably many CCEs.

<sup>19</sup>This claim is also supported by the simulations of [Asker, Fershtman and Pakes \(2022\)](#) and [Banchio and Skrzypacz \(2022\)](#). In particular, the latter finds similar recurrent cycles in a  $19 \times 19$  game.

section we argue that uneven relative learning rates are the *primary cause* of spontaneous coupling. We do this by providing a simple condition that guarantees that reinforcers converge on dominant strategies: reinforcers' relative learning rates must be uniform across all actions. Intuitively, even if algorithms are coupled, uniform learning rates allow agents to evaluate deviations correctly, thus leading them to their dominant strategy.

We first need the following technical assumption:

**Assumption A3** (Full-Support Learning). Let  $G_{-i}^t(a)$  be the distribution over actions of all players but  $i$  at time  $t$ . Then, there exists a  $\chi > 0$  and a  $T$  such that for all  $t > T$ ,  $G_{-i}^t(a) \geq \chi$  for all  $i$  and for all  $a \in A_{-i}$ .

Full-Support Learning ensures that sufficient exploration is carried out by all players in the limit. For example, any game where all agents adopt a  $\varepsilon$ -greedy policy satisfies this assumption. More generally, this assumption states that each action profile is played with positive (albeit small) probability in the limit.

**Theorem 3.** Suppose [Assumption A3](#) is satisfied for all players.<sup>20</sup> In any game with a dominant strategy equilibrium, a reinforcer with  $\text{RLR}(a^i) = \text{RLR}(\tilde{a}^i)$  for all  $a^i, \tilde{a}^i \in A_i$  learns the dominant strategy. If the forward limit set of  $\theta$  is a singleton, then  $\theta$  converges to the dominant strategy equilibrium.

This result is quite general, as we make almost no assumption about the opponent's play and [Assumption S](#) is not required: as long as all actions are played with positive probability even in the limit, the reinforcer will learn to play its dominant strategy. If the game is solvable by strict domination, we can even dispense with [Assumption A3](#). Uniform learning rates ensure any reinforcer will learn the dominant strategy for any number of opponents, even if they adopt different learning algorithms, or the same learning algorithm with different parameters. Imposing uniform relative learning rates can be a relatively stringent requirement on the algorithms; we show with two examples how uniform RLR can be achieved.

**Example 2** (Exponentially-Weighted Moving Average). The EWMA algorithm updates its perceptions  $\theta$  as

$$\theta_a^i(k+1) = \theta_a^i(k) + \alpha \mathbf{1}_{a^i(k)=a} (r(a, a^{-i}) - \theta_a^i(k))$$

Suppose that, given the vector  $\theta$ , action  $a$  is taken with probability  $\mathbb{P}(a^i = a, \theta)$ . The relative learning rate of action  $a$  when perceptions are  $\theta$  is thus  $\mathbb{P}(a^i = a, \theta)$  and, unless the probabilities

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<sup>20</sup>We require this assumption because we formulate the Theorem for games solvable by weak dominance. We can instead drop [Assumption A3](#) when the best-response to any of the opponent's strategies is unique.

are all equal, spontaneous coupling occurs.<sup>21</sup> We can achieve uniform relative learning rates by applying Inverse Probability Weighting (IPW) to the estimates. The Exponentially-Weighted Moving IPW Average (EWMWA) updates its estimates  $\theta$  as

$$\theta_a^i(k+1) = \theta_a^i(k) + \alpha \frac{\mathbf{1}_{a^i(k)=a}}{\mathbb{P}(a^i(k)=a, \theta)} (r(a, a^{-i}) - \theta_a^i(k)).$$

It is immediate to verify that EWMWA has uniform relative learning rates. Then, [Theorem 3](#) shows that an EWMWA algorithm converges on the dominant strategy.<sup>22</sup> EWMWA belongs to the class of no-swap-regret algorithms popular in algorithmic game theory (e.g. EXP3), whose empirical strategies have been shown to converge to correlated equilibria (and thus, here, dominant strategies, see e.g. [Foster and Vohra \(1997\)](#)). [Theorem 3](#) proves a strong notion of convergence, known as last-iterate convergence, and interestingly it does so without invoking the no-regret properties of these algorithms.

**Example 3.** Suppose that a Q-learning agent were able to compute counterfactuals. That is, suppose that, after choosing action  $a^i$  in period  $t$ , the algorithmic agent were able to back out  $r_t(\tilde{a}^i, a^{-i})$  for all  $\tilde{a}^i \neq a^i$ . Then, the agent could update all perceptions simultaneously according to their associated counterfactual reward. This procedure is sometimes referred to as synchronous learning, because learning happens at the same rate for all actions.<sup>23</sup> Synchronous updating affects the learning rates: the second factor in  $\alpha_a^i \cdot (\pi_\epsilon(\theta^i(t)))_a$  (see [Example 1](#)) disappears when agents update synchronously. When asymmetric learning rates arise from missing or asymmetric experiments, counterfactual information (i.e., a correct model of the environment) is sufficient to eliminate the asymmetry.

The following corollary generalizes the intuition of [Example 3](#) to all reinforcers.

**Corollary 2.** Under the same assumptions of [Theorem 3](#), a reinforcer who can compute counterfactuals always learns the dominant strategy. If its forward limit set is a singleton, it converges on the dominant strategy.

It is natural that counterfactuals help to learn to play equilibria. In fact, the theory of Nash equilibrium is based on the assumption that agents can compute the payoff that would have obtained had they played a different action, treating the opponents' strategies

<sup>21</sup>The proof of [Theorem 2](#) can be easily specialized to EWMA to find a pseudo-steady-state exhibiting spontaneous coupling.

<sup>22</sup>Notice that EWMA estimates the payoff from each action with a “biased” estimator. Because in every period the algorithm observes only one data point with some probability, a careful econometrician would apply Inverse Probability Weighting (IPW) to inflate the weight for the missing observations.

<sup>23</sup>The term synchronous appears in [Asker, Fershtman and Pakes \(2022\)](#), but the idea of agents learning from counterfactuals is present already in [Tumer and Khani \(2009\)](#).

as fixed. This in turn allows them to evaluate incentives to deviate. [Corollary 2](#) establishes that reinforcer algorithms successfully rule out dominated strategies, provided they have access to a method to compute counterfactuals. Reinforcers with counterfactuals will learn to play the (unique) equilibrium.

We can extend the intuition that uniform learning rates ensure that players correctly estimate the value of deviations to more general settings. This is particularly relevant for pricing games that do not have equilibria in dominant strategies. Let us consider the procedure of iterated elimination of strictly dominated strategies (IESDS), where we restrict deletion to strategies strictly dominated by another *pure* strategy. This is because reinforcers are not designed to learn or play mixed strategies.<sup>24</sup>

**Definition 7.** We say that an action  $a^i \in A_i$  is *pure-rationalizable* if there is an order of IESDS such that  $a^i$  survives the IESDS procedure.

In general, for a certain order of deletion of dominated strategies action  $a^i$  might get eliminated. However, as long as there is an order such that  $a^i$  survives the IESDS process, we consider  $a^i$  pure-rationalizable. Our next theorem shows that reinforcers with access to counterfactuals only play pure-rationalizable strategies in the limit.

**Theorem 4.** *Let all players in game  $G$  learn through a reinforcer using a  $\varepsilon$ -greedy policy with  $RLR(a^i) = RLR(\tilde{a}^i)$  for all  $a^i, \tilde{a}^i \in A_i$  for all  $i \in N$ . Assume  $\varepsilon > 0$  is small enough so that the reward's order is preserved, i.e. if  $\underline{a}^{-i}$  is the preferred profile of actions of agent  $i$ 's opponents,  $\mathbb{E}_{\pi^{-i}}[r(a^i, \underline{a}^{-i})] > \mathbb{E}_{\pi^{-i}}[r(\tilde{a}^i, \underline{a}^{-i})]$  when  $r(a^i, \underline{a}^{-i}) > r(\tilde{a}^i, \underline{a}^{-i})$ . Then, all actions learned by the players are pure-rationalizable in the game  $G$  under the same IESDS order.*

[Theorem 4](#) implies that  $\varepsilon$ -greedy reinforcers with uniform learning rates always learn to play Nash equilibrium strategies in a supermodular game with a unique equilibrium. More generally, in any pure-dominance-solvable game, reinforcers will learn the pure-strategy Nash equilibrium. We can interpret relative learning rates as the relative ability to work out counterfactuals for a given action. Small relative learning rates fail to account for asymmetric play, impairing the ability of the algorithm to best-respond. Uniform learning rates instead guarantee correct counterfactual estimates, so that abandoning dominated strategies is immediate.

**Price Fixing.** Let us now interpret the results in [Asker, Fershtman and Pakes \(2022\)](#) under the light of [Theorems 3](#) and [4](#). The authors simulate algorithmic competition

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<sup>24</sup>[Definition 1](#) rules out that reinforcers can learn the value of randomizing over actions.

in a Bertrand oligopoly, and find that the emergence of collusion depends critically on whether the algorithms rely on synchronous updating.

In the model there are two firms, Alice Inc. and Bob Ltd., with constant marginal costs  $c_A = c_B = 2$ . They sell homogeneous goods and compete by setting prices. The set of feasible prices is composed of 100 equally spaced numbers between 0.01 and 10, inclusive. The set of prices is denoted by  $P = \{p^1, \dots, p^{100}\}$ . Consumers buy from the firm with the lowest price, and the demand function is

$$d_i(p_i, p_{-i}) = \begin{cases} 1 & \text{if } p_i < p_{-i} \text{ and } p_i \leq 10 \\ \frac{1}{2} & \text{if } p_i = p_{-i} \text{ and } p_i \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

As the authors note, there are two Nash equilibria of this game, one ( $E_1$ ) with  $p_A = p_B = 2.0282$  and one ( $E_2$ ) with  $p_A = p_B = 2.1291$ . The multiplicity is a consequence of the discretization of the price space. [Asker, Fershtman and Pakes \(2022\)](#) assumes that firms use a Q-learning with zero discount factor and they consider two variations, both of which are *greedy*, i.e., the action taken is always the one with the highest estimated value.<sup>25</sup>

- (i) Asynchronous Q-learning: the algorithm updates only the perception of the action taken in each period;
- (ii) Synchronous Q-learning: the algorithm updates all perceptions in each period, with the profit that it could have obtained had he played the other action instead, but holding the opponent's action fixed.

Under asynchronous learning the relative learning rate of a price is either 0 or 1.<sup>26</sup> In this case we expect spontaneous coupling to occur, and indeed [Asker, Fershtman and Pakes \(2022\)](#) finds supra-competitive prices. Instead, by construction then synchronous greedy Q-learning is a reinforcer with uniform relative learning rates (in fact, it fits exactly [Example 3](#)). Applying [Theorem 4](#), which allows us to deal with settings where there is no dominant strategy, we deduce that, if the firms employ synchronous learning, they can only converge to competitive pricing. Moreover, the following corollary also suggests that the parameters of the algorithms are irrelevant for the convergence result.

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<sup>25</sup>With this specification greedy Q-learning becomes equivalent to the EWMA algorithm of [Example 2](#) with a greedy policy.

<sup>26</sup>Notice that, because the algorithms considered here are *greedy*, there is no exploration. In particular, the relative learning rates are not  $\frac{\epsilon}{2}$  and  $1 - \frac{\epsilon}{2}$  but 0 and 1, because the algorithms updates only the action that was taken in the last period. Technically, this means that [Assumption A3](#) is not satisfied. Nonetheless we can apply our results as long as sufficient exploration is carried out, in this case via optimistic initialization.

**Corollary 3.** *In the Bertrand oligopoly game above, if players adopt any  $\varepsilon$ -greedy separable reinforcers with a small  $\varepsilon > 0$  such that the relative speed of learning is the same across all prices, the reinforcers will learn to play either  $p_1 = 2.0282$  or  $p_2 = 2.1291$ .*

*Proof.* [Theorem 4](#) guarantees that two  $\varepsilon$ -greedy reinforcers learn a pure-rationalizable strategy. Discretized homogeneous Bertrand games have only two pure-rationalizable strategies, the two lowest prices above marginal cost, which are also the Nash Strategies.  $\square$

## 5 Application: Learning-Robust Mechanism Design

The previous sections demonstrated that reinforcers can fail to learn to play their dominant strategy. This has profound implications for market and mechanism design, often reliant on dominant strategy implementations. However, we can leverage [Theorem 3](#) and [Corollary 2](#) to alleviate the problem. We propose designing counterfactual feedback to prevent spontaneous coupling. Effectively, this means designing a dominant strategy mechanism with a feedback rule that guarantees that reinforcers learn the dominant strategy. Moreover, we seek the minimal feedback necessary to accomplish this goal, to avoid the unintended consequences of excessive information revelation. All formal statements and proofs in this section are presented in [Appendix E](#).

Consider a canonical model of implementation with private information. Let  $\mathcal{X}$  be the set of possible outcomes, and let there be a set  $N$  of agents with types  $(\lambda^i)_{i \in N} \in \times_{i \in N} \Lambda_i = \Lambda$  fixed over time.<sup>27</sup> Type  $\lambda^i$  determines agent  $i$ 's preferences  $u^i: \mathcal{X} \times \Lambda_i \rightarrow \mathbb{R}$  over outcomes. A direct revelation mechanism requires each agent to report a type  $\hat{\lambda}^i$ . The mechanism then maps the reported type profile  $\hat{\lambda}$  to an outcome,  $f(\hat{\lambda})$ . We say a mechanism is *strategy-proof* if it is a direct revelation mechanism and reporting truthfully is a dominant strategy:

$$u^i(f(\lambda^i, \lambda^{-i}), \lambda^i) \geq u^i(f(\hat{\lambda}^i, \lambda^{-i}), \lambda^i) \quad \forall \hat{\lambda}^i \neq \lambda^i.$$

Assume further that a subset of agents  $L \subseteq N$  act according to the choices of their own reinforcer. Agents in  $L$ , learners, assess the value of each individual report over multiple iterations of the mechanism. Agents in  $N \setminus L$  are instead rational and myopic. That is, they rationally play their static dominant strategy instead of trying to manipulate the learning agents. In each period both rational and learning agents choose a report. Once

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<sup>27</sup>It is simple to extend this result to allow for types drawn i.i.d. in every period, but for simplicity we stick to a constant type in this section.

the mechanism selects the outcome, payoffs realize and learners update their estimates. We call this setting a *hybrid market*, because rational and learning agents coexist.

If the mechanism is strategy-proof, myopic rational agents play their dominant strategy and report truthfully. However, as we have seen in [Sections 3 and 4](#) coupling between independent algorithms may lead to behavior different from dominant strategy, consistent with collusive agreements, so that simple strategy-proof mechanisms may fail the designer. We thus seek *learning-robust strategy-proof mechanisms* (LRSMs).

**Definition 8.** Suppose agents in  $L$  adopt a (separable) reinforcer. A strategy-proof mechanism is *learning-robust* if each agent  $l \in L$  learns the truthful report  $\lambda_{\text{truthful}}^l$ .

We assume that the designer can reveal information to the participants of the mechanism ex-post, and that such information is used by the algorithms to make inference about payoffs. In particular, for each agent the designer can choose a partition of the space  $\Lambda_{-i}$  of opponents' types, and after observing the reports, reveal the element of the partition that contains the realized type profile. We refer to this ex-post revelation as *feedback provision*. Of course, in order to apply our results in [Section 4](#), feedback needs to be sufficiently informative, namely it must allow full counterfactual evaluation. For any given strategy-proof mechanism  $f$ , we then look for a LRSM for  $f$ , that is, an ex-post feedback policy which allows agents to evaluate their counterfactuals.

The designer can always opt for a full revelation feedback policy.<sup>28</sup> Revealing everyone's report after the mechanism yields a LRSM, but it may have unintended consequences: insisting on revealing all private information may facilitate tacit or explicit collusion, and it may result in large communication costs. We then turn our attention to LRSM with reduced communication burden. Formally, we say that a feedback policy is *more private* than another one if the former is a coarser partition of the type space than the latter. This (partial) order is a lattice, and thus there exist both a minimally- and a maximally-private feedback policy. The full revelation policy is minimally private. Instead, the maximally private policy communicates just enough information to ensure that each agent can just compute its counterfactuals. Such feedback is informationally-equivalent to the well-known *menu* formulation of the mechanism ([Hammond, 1979](#)).

**Theorem 5.** Let  $f$  be a strategy-proof mechanism. Then,

1. There always exists a LRSM for  $f$ ,
2. The maximally-private LRSM for  $f$  is a menu description.

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<sup>28</sup>The full revelation policy is the finest partition of  $\Lambda_{-i}$ .

Menu descriptions are the ex-post feedback counterpart of menu mechanisms. [Hammond \(1979\)](#) defines menu mechanism as providing each agent with a menu, which depends on the profile of reports of the opponents, and let the agent choose his preferred outcome. Similarly, maximally private feedback is also a menu: after receiving all reports and implementing the outcome, the designer communicates a menu, dependent on the profile of reports of the opponents, that maps each type to the corresponding outcome clinched by the agent reporting such type.

The feedback of menu descriptions aggregates market information, which helps agents evaluate the true value of truthful reporting.<sup>29</sup> [Parkes \(2004\)](#) argues that mechanism design can play an important role in shaping algorithmic systems. The author describes *learnable* mechanism design — explicitly designing mechanisms to improve performance considering the agent’s adaptive behavior. As he suggests, “*a useful learnable mechanism would provide information, for example via price signals, to maximize the effectiveness with which individual agents can learn equilibrium strategies*”. We formalize this intuition by showing that feedback design can make traditional strategyproof settings robust to adaptive algorithmic players by providing price signals, or menu descriptions.

While minimal, menu descriptions may become unwieldy for complex mechanisms, such as multi-unit VCG auctions. Nonetheless, our notion remains useful for simpler cases. For example, note that online ad auctions have partly begun to provide feedback to their participants. Google’s auctions for display advertising provide “minimum bid to win” information after each auction has concluded to all participants. The minimum bid to win is indeed the menu description for a single-item allocation problem.

## 6 Conclusion

This paper analyzes collusion in games played by “online” learning algorithms, i.e. algorithms that are updated and take decisions after each step. We take a theoretical perspective and, complementing burgeoning empirical and numerical evidence, we identify a new driver of collusive behavior specific to algorithmic players. We first address the issue of analytical intractability of strategic interaction among algorithms by showing that it can be approximated with a dynamical system. Then we apply this framework to dominant-strategy games, and we show that  $(\epsilon)$ -greedy algorithms can learn to collude. We identify the mechanism sustaining collusion, a statistical linkage we call *spontaneous*

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<sup>29</sup>Recently, [Gonczarowski, Heffetz and Thomas \(2023\)](#) discusses the simplicity properties of menu descriptions in experiments with human participants. Here, we argue that menu descriptions may facilitate strategy-proof play from algorithmic participants as well.

*coupling*: when algorithms are slow to realize the value of the competitive action, joint collusion appears more attractive. Involuntary coupling yields self-fulfilling biases in the estimates: we demonstrate this intuition in a Prisoner’s Dilemma with Q-learning agents. We expect the techniques developed to analyze the simple Prisoner’s Dilemma to yield insight in games with more complex strategic structure. In particular, we believe similar techniques can help understanding how AI algorithms reach tit-for-tat strategies when given monitoring technology.

Finally, we design mechanisms that are robust to the presence of algorithmic agents. The ideas we outline are sensible design principles even when dealing with algorithms that are not reinforcers. For example, regret-minimizing algorithms obtain better guarantees when provided with counterfactual information. Naturally, in the spirit of [Abreu and Matsushima \(1992\)](#), there may be scope to enlarge the set of implementable outcomes, due to the bounded-rationality nature of our algorithmic agents. This of course may benefit the market designer and possibly the consumers. A full characterization of the implementable set is out of scope for this paper but of independent interest.

An interesting research direction is to study the regulation of algorithmic competition. Such program presents some challenges. For one, this paper highlights the limited role that monitoring plays in the emergence of spontaneous coupling. The results and intuitions built upon reward-punishment schemes may be ill-suited to limit the extent of algorithmic collusion. More generally, we make some progress towards answering questions about the differences between algorithmic and traditional markets, but it remains an open question whether known regulatory tools apply to these settings, and what other unique features of algorithmic competition deserve separate attention.

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## Appendix A Proofs

We restate [Theorem 1](#) in its more formal version:

Let  $(\theta, \pi)$  be a collection of reinforcers that satisfy [Assumption A1](#) individually, and let the domain  $\mathcal{T} \subset \mathbb{R}^{\sum_i d_i}$  of  $\theta$  be a compact set. Let  $(H_j)_{j \in J}$  be the collection of  $\theta$ 's maximal Lipschitz-continuity domains, that is, let  $H_j$  be the largest open set such that  $\theta$  is Lipschitz over  $H_j$  and there is a discontinuity on  $\partial H_j$ . For all  $j \in J$  the collection of Cauchy problems

$$\begin{cases} \frac{d\Theta^i(t)}{dt} = \alpha \mathbb{E}_{\pi^i, \pi^{-i}} [D^i(\pi^i, r(\pi^i, \pi^{-i}), \Theta^i(t))] \\ \Theta^i(0) = y_0^i \end{cases}$$

has a solution  $\Theta^i$  for all  $i$  over  $H_j$  for all  $y_0 \in H_j$ . There exists a sequence of processes  $\{\theta^n\}_{n \in \mathbb{N}}$  such that:

- $\mathbb{E}[\theta^n(\tau(k))] = \mathbb{E}[\theta(k)]$  for all  $k$ ,  $\tau(k) = \inf\{t \mid \theta^n(t) \text{ jumped } k \text{ times}\}$ ,
- the infinitesimal generators  $\mathcal{A}D_n(\theta)$  are all identical to  $\mathcal{A}D_1(\theta)$  for all  $\theta \in H_j$  and  $n$ ,
- $\lim_{n \rightarrow \infty} P\left\{\sup_{t \leq T} \|\theta^n(t) - \theta(t)\| > \eta\right\} = 0$  for all  $T \geq 0$  and  $\eta > 0$  such that  $\{\theta(t)\}_{t \leq T} \subset H_j$ .

**Proof of [Theorem 1](#).** The existence of a solution for the Cauchy problem is guaranteed by [Assumption A1](#) and the restriction to the maximal continuity domains  $H_j$ . In particular, notice that one can write  $D^i(\pi^i(\theta), r(\pi^i(\theta), \pi^{-i}(\theta^{-i}), \theta^i))$  as a map  $\hat{D}(\theta, Y)$  where  $Y$  is a random variable representing the uncertainty introduced by the policies  $\pi$ .

We can divide the proof of [Theorem 1](#) in two main steps:

1. Finding the correct continuous-time embedding of the reinforcer  $\theta$ ;
2. Identifying a scaling that guarantees limits are well-defined.

First, let us fix a compact ball of radius  $r$  in  $\mathbb{R}^{\sum_i d_i}$ . We will consider the set  $E = H \cap B(r)$  with the Borel intersection sigma algebra. Since we can choose  $r$  to be as large as we want, the approximation will hold for any finite values of  $\theta$ . Let us add one component to the vector  $\theta$ , in position  $\sum_i d_i + 1$ , which will keep track of the iteration  $k$ .

As far as the first step is concerned, let us define a Poisson process  $N_1$  of rate  $\lambda_1 = 1$ . Consider the sequence of (stochastic) arrival times  $0 < \tau_1 < \tau_2 < \tau_3 < \dots$ . We define the process  $\theta^1(t)$  as

$$\theta^1(t) = \theta(k) \quad \text{if } \tau_{k+1} > t \geq \tau_k$$

for all times  $t \geq 0$ . The process  ${}^1\theta(t)$  is a compound Poisson process, cadlag and Markov. Naturally, its  $\sum_i d_i + 1$  component always coincides with the iteration  $k$ . At arrival times the algorithm is equal to its continuous time equivalent  ${}^1\theta$ , which proves item 1 of the Theorem.

We now want to increase the pace of the updates while retaining the same uncertainty in expectation. Intuitively, we can “speed up” the Poisson arrivals, but we also need to “dampen” the jumps accordingly, otherwise the process will diverge to infinity. Formally, we consider a sequence of Continuous-Time Markov Chains indexed by  $n \in \mathbb{N}$  as follows:

- The jump rate  $\lambda_n$  is defined as  $\lambda_n = n$ .
- At each jump, the update in the first  $\sum_i d_i$  components is<sup>30</sup>

$${}^n\theta(t) - {}^n\theta(t^-) = \frac{\alpha}{n} \hat{D}({}^n\theta(t^-), Y)$$

- At each jump, the update in the coordinate  $\sum_i d_i + 1$  is

$${}^n\theta(t) - {}^n\theta(t^-) = \frac{\alpha}{n}$$

Intuitively, the updates of the original process  $\theta$  are scaled down by a factor  $n$  and the last coordinate keeps track of how many updates have occurred scaled by  $n$ .

Consider then the probability measure  $\mu^n(x, dz)$  of the size of the updates starting at  $x$ , with

$$\mu_n(x, dz) = \mathbb{P}\left\{{}^n\theta(\tau^n) \in dz \mid {}^n\theta(0) = x\right\}$$

where  $\tau^n$  is the first exit time of  ${}^n\theta$  from  $x$ . We define the component-wise function

$$F_n(x)^m = \lambda_n \int (z^m - x^m) \mu_n(x, dz), \quad (5)$$

which intuitively describes the expected jump of  ${}^n\theta$  from  $x$  along the  $m$ -th component over one unit of time. In fact,  $F_n$  can be rewritten as

$$F_n(x)^m = n \int \frac{\alpha}{n} \hat{D}^m(x, Y) \mu = \int \alpha \hat{D}^m(x, Y) \mu.$$

We chose the scaling in such a way that the function  $F_n$  is independent of  $n$ . The function  $F_n$  is exactly the infinitesimal generator of the compound Poisson process  ${}^n\theta(t)$ , therefore item (2) of the Theorem is proved.

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<sup>30</sup>Note that we omit the dependence on the iteration since iterations are now part of the process  ${}^n\theta$ .

Let  $F(x) := \lim_{n \rightarrow \infty} F_n(x)$ . It is clear that  $F(x) = F_n(x)$  for all  $n$ . Moreover the function  $F(x)$  is Lipschitz in every component. We will need a technical lemma:

**Lemma 1.** *Let  $E$  be a compact set in  $\mathbb{R}^{|I| \times |S| \times |A|}$ . There exists a sequence  $\{\varepsilon_n\}_n > 0$  with  $\lim_{n \rightarrow \infty} \varepsilon_n = 0$  such that*

$$\lim_{n \rightarrow \infty} \sup_{x \in E} \lambda_n \int_{|z-x| > \varepsilon_n} |z-x| \mu_n(x, dz) = 0$$

Moreover,

$$\sup_n \sup_{x \in E} \lambda_n \int_E |z-x| \mu_n(x, dz) < \infty$$

*Proof.* Since  $E$  is compact  $\hat{D}(\theta, Y)$  must be bounded. Let  $M = \sup_{\theta \in E} \hat{D}(\theta, Y)$  across dimension, and note that  $M < +\infty$ . Let then  $\{\frac{M}{n}\}_n$  be a sequence satisfying the assumptions of the Lemma, and notice that  $\int_{|z-x| > \varepsilon_n} |z-x| \mu_n(x, dz) = 0$  for all  $x$  and  $n$ . We thus proved the first claim. The second claim follows a simple observation:  $|z-x| \mu_n(x, dz) \leq \frac{M}{n}$  for all  $x$ . Since  $\lambda_n = n$ , we obtain that

$$\sup_n \sup_{x \in E} \lambda_n \int_E |z-x| \mu_n(x, dz) = M < \infty$$

which concludes the proof.  $\square$

This lemma verifies one of the conditions of the following Theorem taken from [Kurtz \(1970\)](#):

**Theorem 2.11.** *Suppose there exists  $E \subset \mathbb{R}^k$ , a function  $F: E \rightarrow \mathbb{R}^k$  and a constant  $M$  such that  $|F(x) - F(y)| \leq M|x - y|$  for all  $x, y \in E$  and*

$$\lim_{n \rightarrow \infty} \sup_{x \in E_n \cup E} |F_n(x) - F(x)| = 0$$

Let  $X(t, x_0), 0 \leq t \leq T, x_0 \in E$  satisfy

$$X(0, x_0) = x_0, \quad \dot{X}(t, x_0) = F(X(t, x_0))$$

Suppose additionally that the sequence  $F_n$  satisfies the conditions of [Lemma 1](#), then for every  $\eta > 0$

$$\lim_{n \rightarrow \infty} P\left\{\sup_{t \leq T} |X_n(t) - X(t, x_0)| \geq \eta\right\} = 0$$

If  $X(t, x_0) = \Theta$ , we can verify that the assumptions of the Theorem hold:

- since  $\hat{D}$  is Lipschitz, and  $F_n = F$  are integrals of Lipschitz functions, it is clear that  $|F(x) - F(y)| \leq M|x - y|$  holds,
- $\lim_n \sup_{x \in H} |F_n(x, t) - F(x, t)| = 0$  is satisfied by definition of  $F_n = F$ ,
- the conditions of [Lemma 1](#) are verified.

Then, Theorem 2.11 implies that for every  $\eta > 0$

$$\lim_{n \rightarrow \infty} P \left\{ \sup_{t \leq T} |^n \theta(t) - \Theta(t)| \geq \eta \right\} = 0$$

which proves item 3 of the Theorem and concludes the proof.

As advanced at the beginning, the process  $\Theta$  is a deterministic process with all uncertainty collapsed into the drift component.  $\square$

**Proof of Proposition 2.** This result relies on the following theorem from [Filippov \(1988\)](#):

**Theorem 1, Chapter 2, Section 7.** *Let  $S$  be a compact domain. Let  $G(t, x)$  be a nonempty, bounded, closed, convex set-valued function that is upper semicontinuous in  $t, x$  for all  $(t, x) \in S$ . Then for any point  $(t_0, x_0) \in S$  there exists a solution of the problem*

$$\dot{x} \in G(t, x), \quad x(t_0) = x_0$$

*and if the domain  $S$  contains a cylinder  $Z(t_0 \leq t \leq t_0 + a, |x - x_0| \leq b)$ , the solution exists at least on the interval*

$$t_0 \leq t \leq t_0 + d, \quad d = \min \left\{ a; \frac{b}{m} \right\} \quad m = \sup_Z |G(t, x)|$$

Let  $G(t, x)$  be defined as the vector field given in the statement of [Proposition 2](#), and notice that  $G$  is time-invariant; also let  $S$  be a compact ball in  $\mathbb{R} \times \mathbb{R}^{|Z| \times |\mathcal{A}|}$ . We show that  $G$  satisfies all the conditions:

- it is nonempty over  $\mathbb{R} \times \mathbb{R}^{|Z| \times |\mathcal{A}|}$ ,
- it is bounded everywhere, since each individual  $F_a$  is bounded over  $\omega_{a,a} \cap S$ ,
- it is closed and convex, as it is clear from the definition above,
- it is upper semicontinuous: to see this, select a convergent sequence in the domain. If the sequence is entirely contained in a  $\omega_j$ , then continuity is clear. Instead, sup-

pose that the sequence lies in an  $\omega_{a,a}$  but its limit lies on  $\partial\overline{\omega}_{a,a}$ . Upper semicontinuity is guaranteed by the definition of  $G$  as a convex combination of  $F$  over the overlapping boundaries.

Additionally, note that the domain is any compact ball, therefore we can find a  $S$  that contains any cylinder  $Z$  and a solution to this differential inclusion is global within any compact subset of  $\mathbb{R}^{|I| \times |A|}$ .  $\square$

**Proof of Proposition 1** The existence of the equilibrium  $q_D^{eq}$  follows directly from setting the field over  $\omega_{D,D}$  to zero.

We prove existence of  $q_C^{eq}$  and its related property for one agent; by symmetry, the other agent's Q-values enjoy the same properties. The boundary is defined as  $\Sigma = \{q \in \mathbb{R}^2 : c \cdot q = 0\}$  where  $c = (1, -1)$  and  $\cdot$  denotes the usual dot product. Using the Filippov convention, we can further divide  $\Sigma$  in three regions:

- a *crossing region*,  $\Sigma^c = \{q : (c \cdot (A_C q + b_C))(c \cdot (A_D q + b_D)) > 0\}$
- a *repulsive region*,  $\Sigma^r = \{q : c \cdot (A_C q + b_C) > 0, c \cdot (A_D q + b_D) < 0\}$
- a *sliding region*,  $\Sigma^s = \{q : c \cdot (A_C q + b_C) < 0, c \cdot (A_D q + b_D) > 0\}$

where we have

$$A_C = \begin{bmatrix} \alpha \left(1 - \frac{\varepsilon}{2}\right)(\gamma - 1) & 0 \\ \alpha \gamma \frac{\varepsilon}{2} & -\alpha \frac{\varepsilon}{2} \end{bmatrix} \quad A_D = \begin{bmatrix} -\alpha \frac{\varepsilon}{2} & \alpha \gamma \frac{\varepsilon}{2} \\ 0 & \alpha \left(1 - \frac{\varepsilon}{2}\right)(\gamma - 1) \end{bmatrix},$$

and

$$b_C = \begin{bmatrix} \alpha \left(1 - \frac{\varepsilon}{2}\right) \left(2 - \frac{\varepsilon}{2}\right) g \\ \alpha \frac{\varepsilon}{2} \left(2 + g - g \frac{\varepsilon}{2}\right) \end{bmatrix} \quad b_D = \begin{bmatrix} \alpha \left(1 + \frac{\varepsilon}{2}\right) \frac{\varepsilon}{2} g \\ \alpha \left(1 - \frac{\varepsilon}{2}\right) \left(2 + \frac{\varepsilon}{2} g\right) \end{bmatrix}.$$

We can define the sliding solution as the field  $\frac{d\mathbf{Q}}{dt} = F^s(\mathbf{Q})$  over the sliding region where

$$F^s(\mathbf{Q}) = \frac{c \cdot (A_D \mathbf{Q} + b_D)(A_C \mathbf{Q} + b_C) - c \cdot (A_C \mathbf{Q} + b_C)(A_D \mathbf{Q} + b_D)}{c \cdot (A_D \mathbf{Q} + b_D) - c \cdot (A_C \mathbf{Q} + b_C)}$$

The relative time spent on  $\omega_{C,C}$  at point  $\mathbf{Q}$  is defined as

$$\tau_c = \frac{c \cdot (A_D \mathbf{Q} + b_D)}{c \cdot (A_D \mathbf{Q} + b_D) - c \cdot (A_C \mathbf{Q} + b_C)}$$

The sliding vector field becomes

$$\frac{d\mathbf{Q}_j}{dt} = \frac{\alpha\left(\frac{1}{2}\varepsilon g(2-\varepsilon)(g-1) + (2g + (\gamma-1)\mathbf{Q}_j)(2 + (\gamma-1)\mathbf{Q}_j)\right)}{2(1 + g + (\gamma-1)\mathbf{Q}_j)}$$

for every direction  $j$ . We now search for the point on the surface that sets the field equal to zero; however, note that we must restrict the search to points on  $\Sigma^s$ , because  $\Sigma^s$  is the only region of the boundary where the sliding vector field is well-defined.<sup>31</sup> By setting the field equal to zero and solving for  $\mathbf{Q}_j$ , we find that there is an equilibrium at

$$q_{C,j}^{eq} = \frac{1 + g + \sqrt{(g-1)(g-1 - \varepsilon g + \frac{\varepsilon^2 g}{2})}}{(\gamma-1)}$$

for all  $j$ . This equation has a feasible solution for all  $\varepsilon < 1 - \sqrt{\frac{2-g}{g}}$ .

Finally, we prove our claim regarding the attraction region  $R_C$ . First, notice that any flow whose initial condition is in  $\omega_{C,C}$  necessarily converges to  $q_C^{eq}$ . In fact, since the switching surface  $\bar{\omega}_{C,C} \cap \bar{\omega}_{D,D}$  is either sliding or repulsive from  $\omega_{C,C}$ , any orbits starting in  $\bar{\omega}_{C,C}$  remains in  $\bar{\omega}_{C,C}$ . Then, because the ODE describing the motion in  $\omega_{C,C}$  is linear and its associated eigenvalues are negative, the flow either converges to  $q_C^{eq}$  from  $\omega_{C,C}$ , or hits the sliding portion of the switching surface and *then* converges to  $q_C^{eq}$ . No other convergence is possible because there are no steady states of  $F_{C,C}$  in  $\omega_C$ .

Consider now a flow initialized in  $\omega_{D,D}$ : because  $q_D^{eq} \in \omega_{D,D}$  and  $q_D^{eq}$  would be globally asymptotically stable if there was no switching surface, the flow either converges to  $q_D^{eq}$  or hits the surface before it can converge there. Upon hitting the switching surface, a orbits may either slide onto  $q_C^{eq}$  (if it hits a sliding portion of the surface) or cross into  $\omega_{C,C}$ , after which it finally converges to  $q_C^{eq}$ . It then follows that  $R_C \cap \omega_{D,D}$  is characterized as the set of initial conditions whose flow hits the switching surface at some time  $t < \infty$ . We fully characterize this set. Notice that because the off-diagonal terms of  $A_{D,D}$  are non-negative the trajectories of the systems are ordered: given two initial conditions  $Q, \tilde{Q}$  with  $Q \geq \tilde{Q}$  in the usual componentwise order, for any time  $t$  the trajectories will maintain that order.<sup>32</sup> Consider then the point  $\bar{Q} = (\bar{x}, \bar{x})$ , with  $\bar{x}$  defined as the smallest  $x$  such that  $(x, x)$  is crossing on  $\bar{\omega}_{C,C} \cap \bar{\omega}_{D,D}$ , i.e.,

$$\bar{x} = \min\{x : \hat{\mathbf{N}} \cdot F_C(x, x) \geq 0, \hat{\mathbf{N}} \cdot F_D(x, x) \geq 0\}. \quad (6)$$

<sup>31</sup>Note that without this additional condition, there are two stationary points of the vector field above.

<sup>32</sup>Dynamical systems with non-negative off-diagonal terms in their Jacobian are termed *cooperative*, and enjoy many “nice” properties; see, e.g., [Hirsch \(1982\)](#).

We claim that, if one solves the ODE backwards in time with initial condition  $\bar{Q}$ , the image of the flow of this solution traces the boundary between  $R_D \cap \omega_{D,D}$  and  $R_C \cap \omega_{D,D}$ . Denote by  $B_C$  the set of points in  $\mathbb{R}^2$  that belong to such orbit. Because of the ordering of solutions, any flow with initial condition  $(Q_D(0), Q_C(0)) \in \omega_{D,D}$  such that  $Q_C(0) > \bar{Q}_C(Q_D(0))$ , where  $\bar{Q}_C(Q_D(0))$  is such that  $(Q_D(0), \bar{Q}_C(Q_D(0))) \in B_C$ , must also hit the switching surface at some  $(x, x) > (\bar{x}, \bar{x})$ , and therefore either cross or slide and ultimately converge to  $q_C^{eq}$ . Conversely, because  $\bar{x}$  is the “last” crossing point, any flow with  $Q_C(0) < \bar{Q}_C(Q_D(0))$  does not hit the switching surface and therefore converges to  $q_D^{eq}$ , which finally proves that  $B_C$  is the boundary of  $R_C \cap \omega_{D,D}$ .

With this characterization, it is easy to perform comparative statics about the attraction regions, because it amounts to study the comparative statics of  $B_C$  and, thanks to the ordering of solution, this simplifies to perform comparative statics on  $\bar{x}$ . For example, assuming  $\gamma = 0.9$  (as in the figures of Section 3) one easily solves the linear program in Equation (6) above to obtain that

$$\bar{x}(\varepsilon, g) = \frac{20 - 5\varepsilon^2 g - 10\varepsilon}{1 - \varepsilon}$$

It is then easy to see that

$$\begin{aligned} \frac{\partial \bar{x}(\varepsilon, g)}{\partial g} &< 0 \\ \frac{\partial \bar{x}(\varepsilon, g)}{\partial \varepsilon} &> 0 \end{aligned}$$

That is, when the profitability of cooperation increases, the attraction region of defection becomes smaller and that of cooperation expands; similarly, when the exploration rate increases the attraction region of defection expands. □

**Proof of Corollary 1.** The result follows immediately from the proof of Proposition 1. In particular, it is sufficient to compute

$$\tau_c = \frac{c \cdot (A_D \mathbf{Q} + b_D)}{c \cdot (A_D \mathbf{Q} + b_D) - c \cdot (A_C \mathbf{Q} + b_C)}$$

at  $\mathbf{Q} = q_C^{eq}$ . □

**Proof of Theorem 2.** We wish to show that for fixed functions  $U$  and  $V$ , there always exists an open set of relative learning rates  $\{\alpha_j(\omega_{k,k})\}_{j,k=C,D}$  such that spontaneous coupling occurs. That is, we seek an open set  $A \subset \mathbb{R}_+^4$  of relative learning rates such that

- (i) If each agent learns through the the same greedy reinforcer defined by  $U$ ,  $V$ , and  $\{\alpha_j(\omega_{k,k})\}_{j,k=C,D}$ ;
- (ii) If Assumption S is satisfied;

for all  $\{\alpha_j(\omega_{k,k})\}_{j,k=C,D} \in A$  there exists a pseudo-steady-state on the boundary  $\bar{\omega}_{C,C} \cap \bar{\omega}_{D,D}$ , for a dynamical system such that

$$\begin{cases} \dot{\theta}^C = \alpha_C [U(\theta^C, r(C, C)) + V(\theta)] \\ \dot{\theta}^D = (1 - \alpha_C) [U(\theta^D, r(D, C)) + V(\theta)] \end{cases}$$

in the half-plane where  $C$  is the preferred action, and

$$\begin{cases} \dot{\theta}^C = (1 - \alpha_D) [U(\theta^C, r(C, D)) + V(\theta)] \\ \dot{\theta}^D = \alpha_D [U(\theta^D, r(D, D)) + V(\theta)] \end{cases}$$

in the half-plane where  $D$  is the preferred action. In the above, we write  $\alpha_C = \alpha_C(\omega_{C,C})$ ,  $\alpha_D = \alpha_D(\omega_{D,D})$ , and assume without loss of generality that learning rates sum to 1.<sup>33</sup>

For given  $U$ ,  $V$  and  $\alpha$ 's, a pseudo-steady-state  $\theta^*$  exists if and only if there are  $(\theta^*, \tau^*)$  such that

$$\begin{cases} \alpha_C \tau^* U(\theta^*, r(C, C)) + (1 - \alpha_D)(1 - \tau^*) U(\theta^*, r(C, D)) + V(\theta^*) [\alpha_C \tau^* + (1 - \alpha_D)(1 - \tau^*)] = 0 \\ (1 - \alpha_C) \tau^* U(\theta^*, r(D, C)) + \alpha_D (1 - \tau^*) U(\theta^*, r(D, D)) + V(\theta^*) [\tau^* (1 - \alpha_C) + \alpha_D (1 - \tau^*)] = 0. \end{cases} \quad (7)$$

and the dynamical system describes a sliding boundary.

Our first step is to show that, in every Prisoner's Dilemma, for every  $\tau \in [0, 1]$  there exist  $\alpha_C = \alpha_D = \alpha^\tau \in [0, 1]$  and  $\theta(\tau)$  such that Equation (7) is satisfied. Therefore, take  $\tau$  as given until otherwise stated, and we search for such  $\alpha^\tau$  and  $\theta(\tau)$ . We can rewrite Equation (7) as seeking a convex combination of two vectors that results in the zero vector:

$$\alpha \vec{x} + (1 - \alpha) \vec{y} = \vec{0}$$

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<sup>33</sup>Note that we also abuse notation and write  $V(\theta)$  when the domain of  $V$  is  $\mathbb{R}^2$ . Since we carry out the whole proof on the diagonal where  $\theta_C = \theta_D$ , there is no loss.

where

$$\vec{x}(\theta) = \begin{bmatrix} \tau [U(\theta, r(C, C)) + V(\theta)] \\ (1 - \tau) [U(\theta, r(D, D)) + V(\theta)] \end{bmatrix} \quad \vec{y}(\theta) = \begin{bmatrix} (1 - \tau) [U(\theta, r(C, D)) + V(\theta)] \\ \tau [U(\theta, r(D, C)) + V(\theta)] \end{bmatrix}$$

Let  $\theta^1$  be the value of  $\theta$  such that  $U(\theta^1, r(C, C)) + V(\theta^1) = 0 \implies V(\theta^1) = -U(\theta^1, r(C, C))$ .

Further let  $\theta^2$  be the value of  $\theta$  such that  $U(\theta^2, r(C, D)) + V(\theta^2) = 0 \implies V(\theta^2) = -U(\theta^2, r(C, D))$ .

We now represent the vectors at  $\theta^1$  and  $\theta^2$  in  $\mathbb{R}^2$  in [Figure 9](#), where the horizontal axis represents the first component and the vertical axis the second component. We have

$$\vec{x}(\theta^1) = \begin{bmatrix} 0 \\ (1 - \tau) [U(\theta^1, r(D, D)) - U(\theta^1, r(C, C))] \end{bmatrix} \quad \vec{y}(\theta^1) = \begin{bmatrix} (1 - \tau) [U(\theta^1, r(C, D)) - U(\theta^1, r(C, C))] \\ \tau [U(\theta^1, r(D, C)) - U(\theta^1, r(C, C))] \end{bmatrix}$$

and

$$\vec{x}(\theta^2) = \begin{bmatrix} \tau [U(\theta^2, r(C, C)) - U(\theta^2, r(C, D))] \\ (1 - \tau) [U(\theta^2, r(D, D)) - U(\theta^2, r(C, D))] \end{bmatrix} \quad \vec{y}(\theta^2) = \begin{bmatrix} 0 \\ \tau [U(\theta^2, r(D, C)) - U(\theta^2, r(C, D))] \end{bmatrix}$$

Notice that the payoffs in a Prisoner's Dilemma are ordered as follows:

$$r(D, C) > r(C, C) > r(D, D) > r(C, D),$$

which, together with  $U$  being increasing in the second argument, implies that the vector are positioned in the plan as represented in [Figure 9](#). Moreover, since according to [Definition 5](#) we have  $\frac{\partial V}{\partial \theta_a^i} < -\frac{\partial U}{\partial \theta_a^i}$ , it also holds that the curves  $\vec{y}(\theta^1 + (\theta^2 - \theta^1)t)$  and  $\vec{x}(\theta^1 + (\theta^2 - \theta^1)t)$  lie on the left and right of the vertical axis, respectively.

Define now  $f: [0, 1] \rightarrow \mathbb{R}^2$  as

$$f(t) = \begin{cases} (1 - 4t)\vec{x}(\theta^1) + 4t\vec{y}(\theta^1) & t \in [0, \frac{1}{4}] \\ \vec{y}(\theta^1 + (\theta^2 - \theta^1)(4t - 1)) & t \in [\frac{1}{4}, \frac{1}{2}] \\ (3 - 4t)\vec{y}(\theta^2) + (4t - 2)\vec{x}(\theta^2) & t \in [\frac{1}{2}, \frac{3}{4}] \\ \vec{x}(\theta^1 + 4(\theta^2 - \theta^1)(1 - t)) & t \in [\frac{3}{4}, 1] \end{cases}$$

Note that, by continuity of  $U$  and  $V$ ,  $f(t)$  is a loop based in  $x(\theta^1)$ . We can show that  $f$

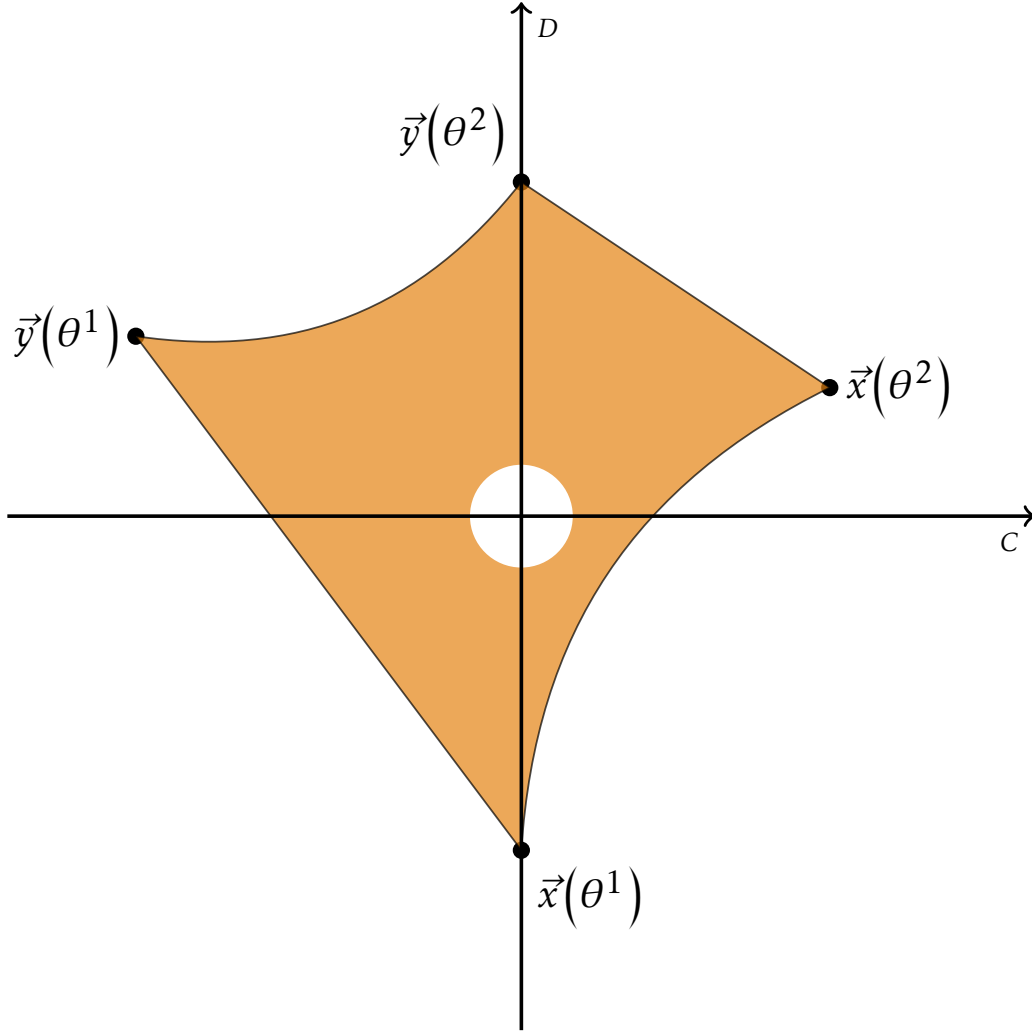


Figure 9: Representation of  $\vec{x}(\theta^1)$ ,  $\vec{x}(\theta^2)$ , and associated homotopical argument. Horizontal axis is labeled as  $C$  (resp.,  $D$ ) because the first (resp., second) component of  $\vec{x}$  and  $\vec{y}$  describes the dynamics of the sliding field for  $\theta_C$  (resp.,  $\theta_D$ ).

is null-homotopic by providing the following simple homotopy  $H: [0, 1] \times [0, 1] \rightarrow \mathbb{R}^2$ .

$$H(t, s) = \begin{cases} (1 - 4ts)\vec{x}(\theta^1) + 4ts\vec{y}(\theta^1) & t \in [0, \frac{1}{4}] \\ (1 - s)\vec{x}(\theta^1 + s(4t - 1)(\theta^2 - \theta^1)) + s\vec{y}(\theta^1 + s(4t - 1)(\theta^2 - \theta^1)) & t \in [\frac{1}{4}, \frac{1}{2}] \\ (1 - s(3 - 4t))\vec{x}((1 - s)\theta^1 + s\theta^2) + s(3 - 4t)\vec{y}((1 - s)\theta^1 + s\theta^2) & t \in [\frac{1}{2}, \frac{3}{4}] \\ \vec{x}(\theta^1 + 4s(\theta^2 - \theta^1)(1 - t)) & t \in [\frac{3}{4}, 1] \end{cases}$$

We verify that  $H$  is a homotopy between  $f(t)$  and the constant path at  $\vec{x}(\theta^1)$ .

- $H(0, s) = H(1, s) = \vec{x}(\theta^1)$
- $H(t, 0) = \vec{x}(\theta^1)$
- $H(t, 1) = f(t)$

Suppose now that there is no pair  $t, s$  such that  $H(t, s) = (0, 0)$ . Then, by continuity it must be the case that there is an open neighborhood  $V \ni (0, 0)$  such that for all points  $z \in V$ ,  $z \notin \text{Im}(H)$ . Note that we can restrict the loop  $f$  to a the domain  $\mathbb{R}^2 \setminus V$ , and because  $V \notin \text{Im}(H)$  we can restrict the homotopy to a homotopy  $H: [0, 1]^2 \rightarrow \mathbb{R}^2 \setminus V$ . Thus we proved that a loop around the open set  $V$  is null-homotopic, which is equivalent to proving that  $\mathbb{R}^2 \setminus V$  is simply connected, a contradiction.

Therefore, there exists a pair  $\bar{t}, \bar{s}$  such that  $H(\bar{t}, \bar{s}) = (0, 0)$ . There is a bijective transformation between  $(t, s)$  and  $(\theta, \alpha)$  over their respective domains, which guarantees that the system of [Equation \(7\)](#) is satisfied.

Notice that we allowed for  $\alpha \in (0, 1)$  so far. However, it makes little sense to allow for  $\alpha < \frac{1}{2}$ : the algorithm would learn about actions it considers suboptimal faster than those it considers best. We observe the following:

$$\frac{1}{2}\tau U(\theta, r(C, C)) + \frac{1}{2}(1 - \tau)U(\theta, r(C, D)) < \frac{1}{2}(1 - \tau)U(\theta, r(D, C)) + \frac{1}{2}\tau U(\theta, r(D, D)),$$

for all  $\tau$  small enough. This inequality follows from the ordering of rewards in a Prisoner's Dilemma game. This in particular implies that the line  $\frac{1}{2}\vec{x}(\theta^1 + (\theta^2 - \theta^1)t) + \frac{1}{2}\vec{y}(\theta^1 + (\theta^2 - \theta^1)t)$  lies above the 45 degree line. Thus, we can restrict the homotopy to the parameter space  $\alpha \in [\frac{1}{2}, 1]$  without affecting the result. This guarantees the existence of a parameter  $\alpha > \frac{1}{2}$  which sets the sliding vector field to zero as long as  $\tau$  is small enough. It then means that for every  $\tau$  small enough there exist  $\alpha^\tau \in [\frac{1}{2}, 1]$  and  $\theta^*(\tau)$  that solve [Equation \(7\)](#). Let this threshold be  $\hat{\tau}$ .

The second step starts by observing that not all solutions to [Equation \(7\)](#) are pseudo-steady-states: we need to verify a sliding condition, i.e., that the normal components of the two vector fields to the switching surface must have opposite sign and must be attractive. In equations, this means that the following system of inequalities needs to be satisfied:

$$\begin{cases} (1 - \alpha^\tau)[U(\theta^*(\tau), r(D, C)) + V(\theta^*(\tau))] - \alpha^\tau[U(\theta^*(\tau), r(C, C)) + V(\theta^*(\tau))] \geq 0 \\ \alpha^\tau[U(\theta^*(\tau), r(D, D)) + V(\theta^*(\tau))] - (1 - \alpha^\tau)[U(\theta^*(\tau), r(C, D)) + V(\theta^*(\tau))] \leq 0. \end{cases} \quad (8)$$

We need a preparatory lemma:

**Lemma 2.** *For any fixed  $\tau$ , the only region of  $\theta$  where [Equation \(7\)](#) can be verified is such that*

$$U(\theta, r(C, C)) + V(\theta) \geq 0 > U(\theta, r(D, D)) + V(\theta)$$

*Proof.* The statement follows immediately from inspection of [Figure 9](#). Since the path  $\vec{y}(\theta^1 + (\theta^2 - \theta^1)t)$  for all  $t$  falls within the second quadrant, any  $\vec{x}(\theta^*)$  satisfying [Equation \(7\)](#) must fall within the fourth quadrant, which directly implies the result.  $\square$

Now, rearranging [Equation \(7\)](#) we derive the following equalities:

$$\begin{cases} -(1 - \alpha^\tau)(1 - \tau)[U(\theta^*(\tau), r(C, D)) + V(\theta^*(\tau))] = \alpha^\tau \tau [U(\theta^*(\tau), r(C, C)) + V(\theta^*(\tau))] \\ -(1 - \alpha^\tau)\tau [U(\theta^*(\tau), r(D, C)) + V(\theta^*(\tau))] = \alpha^\tau (1 - \tau)[U(\theta^*(\tau), r(D, D)) + V(\theta^*(\tau))] \end{cases}$$

Substituting these in [Equation \(8\)](#) and rearranging, we obtain

$$\begin{cases} \alpha^\tau \tau [U(\theta^*(\tau), r(C, C)) + V(\theta^*(\tau))] + \alpha^\tau (1 - \tau)[U(\theta^*(\tau), r(D, D)) + V(\theta^*(\tau))] \leq 0 \\ \alpha^\tau (1 - \tau)[U(\theta^*(\tau), r(D, D)) + V(\theta^*(\tau))] + \alpha^\tau \tau [U(\theta^*(\tau), r(C, C)) + V(\theta^*(\tau))] \leq 0 \end{cases}$$

Thus, only one condition needs to be satisfied to guarantee sliding:

$$(1 - \tau)[U(\theta^*(\tau), r(D, D)) + V(\theta^*(\tau))] + \tau [U(\theta^*(\tau), r(C, C)) + V(\theta^*(\tau))] \leq 0$$

[Lemma 2](#) guarantee that a solution to this inequality exists for  $\tau$  sufficiently close to 0. In particular, there exists a  $\bar{\tau}$  such that for all  $\tau \leq \bar{\tau}$  we obtain sliding. Therefore, putting together step one and two, there exists an open set of parameters  $\{\alpha^\tau \mid \tau \leq \min\{\bar{\tau}, \hat{\tau}\}\}$  such that a psuedo-steady-state exists.

Finally, we can perturb the value of  $\alpha$  on either side of the boundary. The region

derived depends continuously on  $\alpha$ , in particular on  $\alpha_C$  and  $\alpha_D$ . Therefore, the same result holds for small perturbations of  $\alpha$  into different  $\alpha_C$  and  $\alpha_D$ , concluding the proof of the Theorem.  $\square$

**Proof of Theorem 3.** We set to prove that, in the limit, the statistic  $\theta^n$  of the dominant action dominates the statistic  $\theta^i$  of any other non-dominant action. First of all, since  $\alpha^{a_i}$  is identical across actions, the evolution in time of  $\theta$  is shifted by  $V(\theta)$ , but  $V$  won't affect the relative rankings of the actions' estimates. Therefore, we drop  $V$  in the rest of the proof, and we focus on  $U$ .

Regardless of the opponent's policy, the reinforcer in every step observes a return in hindsight for every action. We denote by  $r_n(t)$  the return from playing action  $n$  in period  $t$ , whatever the opponents' actions are. We consider the evolution of the weights pairwise:  $\theta^n$  and  $\theta^i$  for every  $i \neq n$ . By assumption, for any sequence of actions taken by the opponent(s),  $r_n(t) \geq r_i(t)$ . In particular, [Assumption A3](#) guarantees that there exists a  $T > 0$  such that  $r_n(t) > r_i(t)$  for any  $t > T$ . Thus, in the limit the derivative  $\dot{\theta}^n$  strictly dominates  $\dot{\theta}^i$  in each point  $(x, t)$ :  $U(x, r_n(t)) > U(x, r_i(t))$ . In particular, there exists a  $\varepsilon$  such that  $U(x, r_n(t)) > U(x, r_i(t)) + \varepsilon$ .

Suppose now that for some  $t \geq 0$ ,  $\theta^n(t) = \theta^i(t)$ . We prove that it can never be the case that  $\theta^i(T) > \theta^n(T)$  for some  $T > t$ .  $\theta^n(t)$  is a solution to the ODE given by  $\dot{\theta}^n(t) = U(\theta^n(t), r_n(t))$  and

$$U(\theta^n(t), r_n(t)) \geq U(\theta^n(t), r_i(t)) = U(\theta^i(t), r_i(t))$$

Thus,  $\dot{\theta}^i(t) < U(\theta^i(t), r_n(t)) = \dot{\theta}^n(t)$ , which implies that  $\theta^n(t + \Delta) > \theta^i(t + \Delta)$  for  $\Delta$  small enough. Thus, for any  $T > t$  there can only be two cases: either  $\theta^n(T) > \theta^i(T)$ , or  $\theta^n(T) = \theta^i(T)$ ; but we have just shown that if the latter case occurs,  $\theta^i$  will stay below  $\theta^n$  again.

Consider instead the case where  $\theta^i(T) > \theta^n(T)$  for some  $T > 0$ . From [Definition 5](#), we have that

$$U(\theta^i(T), r_i(T)) \leq U(\theta^i(T), r_n(T)) < U(\theta^n(T), r_n(T))$$

We want to show that there exists a  $t > T$  such that  $\theta^n(t) = \theta^i(t)$ . To this end, suppose by contradiction that  $\forall t > T, \theta^i(t) > \theta^n(t)$ . Observe that the previous inequalities imply that

$$\left. \frac{d}{ds} \right|_{s=t} (\theta^i(s) - \theta^n(s)) < 0 \quad \forall t > T. \quad (9)$$

Then,  $(\theta^i(t) - \theta^n(t))$  is a monotone decreasing function of time. Because the algorithm is

bounded, it must be the case that

$$\lim_{t \rightarrow +\infty} (\theta^i(t) - \theta^n(t)) = b \geq 0. \quad (10)$$

From the definition of limit and the monotonicity of the derivative, for any  $\epsilon$  there exists a  $t' > T$  such that, for all  $t > t'$ ,

$$\theta^n(t) + b \leq \theta^i(t) < \theta^n(t) + b + \epsilon.$$

However, strict monotonicity of the reinforcer's update implies that there exists a  $\delta > 0$  such that

$$U(\theta^i(t), r_i(t)) \leq U(\theta^n(t), r_i(t)) < U(\theta^n(t), r_n(t)) - \delta. \quad (11)$$

Note that, by Equation (10) and the monotonicity given by Equation (9), the limit of the derivative of the difference  $\theta^i - \theta^n$  be 0:

$$\lim_{t \rightarrow +\infty} (\dot{\theta}^i(t) - \dot{\theta}^n(t)) = \lim_{t \rightarrow +\infty} (U(\theta^i(t), r_i(t)) - U(\theta^n(t), r_n(t))) = 0$$

This is a contradiction of Equation (11). Then, there must exist a  $t$  such that  $\theta^i(t) = \theta^n(t)$ . From the first part of the proof, this implies that  $\forall t' > t$  we have  $\theta^i(t') \leq \theta^n(t')$ . In summary then, either  $\theta^n(0) \geq \theta^i(0)$ , which implies  $\theta^n(t) \geq \theta^i(t)$  for all  $t > 0$ , or  $\theta^n(0) < \theta^i(0)$ , which implies that there exists  $T$  such that  $\theta^n(t) \geq \theta^i(t)$  for all  $t \geq T$ . This concludes the proof.  $\square$

**Proof of Theorem 4.** The proof is largely based upon Theorem 3. We will show that there is always a  $\mathcal{T}$  such that the actions taken after  $\mathcal{T}$  survive an IESDS procedure.

Let  $a^i$  be a strategy for player  $i$  which is dominated by  $b^i$  in the game  $G$ . With the same argument of the proof of Theorem 3 we can show that it must be the case that there exists a  $\mathcal{T}_1$  such that for all  $t \geq \mathcal{T}_1$  we have  $\theta^{b^i}(t) \geq \theta^{a^i}(t)$ . Notice that if  $\theta^{b^i}(t) = \theta^{a^i}(t)$ , then  $\dot{\theta}^{b^i}(t) > \dot{\theta}^{a^i}(t)$ , which implies that for all  $t' > t$  we have a strict inequality  $\theta^{b^i}(t') > \theta^{a^i}(t')$ .

Therefore, after time  $\mathcal{T}_1$  it is as if the agents were playing in a reduced game  $G^1 = (N, (A_i^1)_i, (u_i)_i)$ , where  $A_i^1 = A_i \setminus \{a^i\}$  and  $A_j^1 = A_j$  for all  $j \neq i$ .<sup>34</sup> We can now apply the same idea to this new reduced game  $G^1$  and eliminate a strictly dominated strategy in this reduced game. While IESDS eliminates strategies “in place”, the algorithms abandon dominated strategies over time, reducing the effective game played. Of course, the com-

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<sup>34</sup>Of course, Assumption A3 guarantees that even action  $a^i$  will be played with some positive probability, but the statement of Theorem 4 guarantees that the probability is small enough to not affect the order of the expected rewards.

ponents of  $\theta$  that correspond to dominated actions keep getting updated, but note that if an action is dominated given a larger subset of opponent's strategies, it is also (weakly) dominated given a smaller subset. Therefore, following the usual differential inequality argument, the  $\theta$  corresponding to a dominated action will always remain below that of actions surviving IESDS. We then define  $\mathcal{T}$  as the largest among the  $\mathcal{T}_k$  that correspond to an action being eliminated, and this concludes the proof.  $\square$

## Appendix B Prisoner's Dilemma without symmetry

In [Section 3](#) we derived a number of insights about Q-learners playing the Prisoner's Dilemma under the assumption that algorithms are initialized at the same value ([Assumption S](#)), which, under the fluid approximation, implies that the algorithms' perceptions always coincide. Our aim with this appendix is twofold: on the one hand, we wish to provide a formal explanation of the reason why [Assumption S](#) is necessary to make analytical progress (which also gives us an opportunity to interpret [Assumption S](#)) and derive some results without the assumption; on the other, we discuss to what extent [Assumption S](#) is restrictive and argue that, while not without loss of generality, our analysis of [Section 3](#) correctly identifies spontaneous coupling as the main driver of coordination between Q-learners. Finally, we leverage these consideration to address the concern that in [Section 3](#) coupling is merely *maintained*; here we aim to show that coupling is also *reached* when algorithms start from different perceptions.

### B.1 Four-dimensional system

Let us take the fluid approximation (without symmetry assumption) as given and consider the system of [Equation \(4\)](#); let  $F_{a,a'}$  denote the vector field in region  $\omega_{a,a'}$  for  $a, a' = C, D$ , which define a four-dimensional differential inclusion. [Proposition 2](#) applies to this dynamical system as well (the proof is identical to the two-dimensional case and therefore omitted), so we know that there exists at least one global solution forward in time to [Equation \(4\)](#). However, we now argue that such solution is not unique; in fact, there are uncountably many solutions.

**Non uniqueness.** Given the  $\varepsilon$ -greedy exploration policy, the discontinuities in the vector fields occur along the lines where Alice and Bob's perceptions coincide. In particular, there exist two switching surfaces, one for Alice ( $\Sigma^A$ ) and one for Bob ( $\Sigma^B$ ), given by:

$$\begin{aligned}\Sigma^A &= \{\mathbf{Q} \in \mathbb{R}^4 : Q_C^A = Q_D^A\} \\ \Sigma^B &= \{\mathbf{Q} \in \mathbb{R}^4 : Q_C^B = Q_D^B\},\end{aligned}$$

which are naturally related to the sets  $\omega_{a,a'}$  by:

$$\begin{aligned}\Sigma^A &= (\overline{\omega}_{C,C} \cap \overline{\omega}_{D,C}) \cup (\overline{\omega}_{C,D} \cap \overline{\omega}_{D,D}) \\ \Sigma^B &= (\overline{\omega}_{C,C} \cap \overline{\omega}_{C,D}) \cup (\overline{\omega}_{D,C} \cap \overline{\omega}_{D,D}).\end{aligned}$$

For the purpose of this section, further define the subsets

$$\begin{aligned}\Sigma_C^A &= \overline{\omega}_{C,C} \cap \overline{\omega}_{D,C} \subset \Sigma^A \\ \Sigma_D^A &= \overline{\omega}_{C,D} \cap \overline{\omega}_{D,D} \subset \Sigma^A,\end{aligned}$$

and similarly for Bob. Consider now a  $\mathbf{Q}$  such that  $\mathbf{Q} \in \Sigma_C^A$  but  $\mathbf{Q} \notin \Sigma^B$ ; in words, Alice’s perceptions value cooperation and defection equally, but Bob strictly prefers cooperation. We can then construct a convex combination of the two vector fields  $F_{C,C}$  and  $F_{D,C}$  satisfying the same conditions as in [Section 3.2](#), which would seem to lead to a unique solution to the differential inclusion. However, a global solution must also account for the following case: consider  $\mathbf{Q} \in \Sigma = \Sigma^A \cap \Sigma^B$  and suppose that we expect to see sliding motion on both  $\Sigma^A$  and  $\Sigma^B$ : we now need to find a convex combination of four vector fields, but we only have two conditions to impose, namely that the resulting vector field be parallel to both  $\Sigma^A$  and  $\Sigma^B$ . The system of equations is underdetermined, because there are four variables (one weight for each vector field) and three equations (two parallel conditions and that weights sum to one); each of the infinitely many solutions to this system of equations is also a solution to the differential inclusion. Notice that all these solutions agree with each other within  $\omega_{a,a'}$ , so they can only differ on the discontinuity sets, which have zero measure. Since we argued that the main insights from interacting Q-learners can be gained by looking at the behavior of the system close to and on the discontinuity surfaces, it is important to understand how this difficulty can be overcome.

**Selecting a solution.** The issue of non-uniqueness of solutions to differential inclusions is well-known (see [di Bernardo et al. \(2008\)](#) and references therein) and is in fact an active area of research. In systems like that of [Section 3.1](#) the switching surface has codimension 1, thus it is always possible to find a unique solution; for switching surfaces of codimension greater than or equal to 2, uniqueness is generally lost and *ad-hoc* criteria must be defined to select a solution. Our approach (imposing [Assumption S](#)) can be conceptualized as a “solution-selection” mechanism, much alike assuming that symmetric players in a game will play a symmetric equilibrium. Alice and Bob are a priori identical, and therefore it appears reasonable to study the dynamics under that assumption first. Moreover, as we shall argue later in this section, symmetry is the *one* assumption that guarantees analytical tractability. Despite the clear advantages of working under [Assumption S](#), assessing that our intuitions can be recovered under different assumptions would provide greater robustness to our results. For this reason, we present now a different solution-selection procedure, due to [Dieci and Lopez \(2011\)](#).

### B.1.1 Bi-linear combinations.

The following approach has been proposed for differential inclusions with switching surfaces of codimension 2 and 3, when the higher codimensional switching surfaces are obtained as intersections of codimension-one surfaces; this is exactly our case since  $\Sigma = \Sigma^A \cap \Sigma^B$ . The intuition behind this approach is to assign a weight  $C^i$  to player  $i$  and then mix the vector fields according to a product of the players' weights. We now provide a formal description.

Let  $\hat{\mathbf{N}}^i$  denote a vector in  $\mathbb{R}^4$  orthogonal to  $\Sigma^i$ , where, without loss of generality, we choose  $\hat{\mathbf{N}}^i$  such that it points towards the region of space where  $Q_C^i > Q_D^i$ . Sliding motion, if any, occurs on  $\Sigma^A$ ,  $\Sigma^B$ , and  $\Sigma$ , provided the vector fields on each side “point” in the right direction. In particular, if the system is on  $\Sigma$ , there might be *double sliding*. Intuitively, double sliding occurs when the vector fields in all four regions “push” the system towards surface  $\Sigma$ . Surface  $\Sigma$  is said to be *nodally* attractive if the following hold on, and in a neighbourhood of,  $\Sigma$ :

$$\begin{aligned} \hat{\mathbf{N}}^A \cdot F_{D,D} &> 0, & \hat{\mathbf{N}}^A \cdot F_{D,C} &> 0, & \hat{\mathbf{N}}^A \cdot F_{C,D} &< 0, & \hat{\mathbf{N}}^A \cdot F_{C,C} &< 0 \\ \hat{\mathbf{N}}^B \cdot F_{D,D} &> 0, & \hat{\mathbf{N}}^B \cdot F_{D,C} &< 0, & \hat{\mathbf{N}}^B \cdot F_{C,D} &> 0, & \hat{\mathbf{N}}^B \cdot F_{C,C} &< 0. \end{aligned} \quad (12)$$

In words, Equation (12) requires each surface  $\Sigma^i$  to be sliding irrespective of the action of the opponent: if the system is initialized in a region of the space where  $\Sigma$  is nodally attractive, it will first slide on either  $\Sigma^i$  towards  $\Sigma$ , where it finally begins the double sliding motion.

Suppose for simplicity that  $\gamma = 0$ .<sup>35</sup> When the system is on  $\Sigma$ ,  $\mathbf{Q} = [q, q, q', q']^T$  for some  $q, q' \in \mathbb{R}$ , and the conditions in Equation (12) become

$$\begin{aligned} q &> \bar{q} = g + 1 + \frac{\frac{\epsilon^2}{2}g + 1 - \epsilon g}{1 - \epsilon} \\ q' &> \bar{q} = g + 1 + \frac{\frac{\epsilon^2}{2}g + 1 - \epsilon g}{1 - \epsilon} \end{aligned}$$

This is consistent with the system under symmetry as seen in Figure 5, which shows that also in the symmetric case the switching surface is sliding only for large enough values of the perceptions. The right-hand sides of the above conditions are the same because, even if Alice's and Bob's perceptions are no longer symmetric, their laws of motion are. In fact,

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<sup>35</sup>The same analysis goes through with  $\gamma > 0$ , but equations are somewhat more involved and less transparent to parse.

$\bar{q}$  can be equivalently characterized as

$$\bar{q} = \max \left\{ q : \exists (a, a') \in \{C, D\}^2 \text{ s.t. } \hat{\mathbf{N}}^A \cdot F_{a,a'}([q, q, x, y]^T) = 0, [q, q, x, y]^T \in \bar{\omega}_{a,a'} \right\}, \quad (13)$$

or, in words, the largest value of the perception such that if Alice were sliding on her switching surface  $\Sigma^A$ , her vector field in at least one of the partitions of the space becomes parallel to  $\Sigma^A$ , i.e.,  $\dot{Q}^A(C) = \dot{Q}^A(D)$ . Since the game is symmetric, we could replace Alice with Bob in Equation (13) and obtain the same result. The structure of payoffs of any Prisoner's Dilemma implies that the  $\omega$  where Alice's vector field becomes parallel to her switching surface is  $\omega_{D,C}$ : it is exactly when she defects most of the time and Bob cooperates that Alice obtains the largest payoff.

Dieci and Lopez (2011) suggests defining the weighted vector field using a bilinear formulation and proves that if the sliding surface satisfies nodal attractivity, the vector field thus obtained is unique. Let  $\mathcal{C}^A : \mathbb{R}^4 \rightarrow [0, 1]$  and  $\mathcal{C}^B : \mathbb{R}^4 \rightarrow [0, 1]$  be some functions of the state of the system: we interpret  $\mathcal{C}^A(\mathbf{Q})$  as the weight associated to any  $F_{C,a'}$ , where for  $a' = C, D$ , and similarly for  $\mathcal{C}^B(\mathbf{Q})$ . Dieci and Lopez (2011) proves that there exist unique functions  $\mathcal{C}^A(\mathbf{Q})$  and  $\mathcal{C}^B(\mathbf{Q})$  such that for every  $\mathbf{Q} \in \Sigma$  satisfying nodal attractivity the following holds:<sup>36</sup>

$$\begin{cases} \mathcal{C}^A \mathcal{C}^B \hat{\mathbf{N}}^A \cdot F_{C,C} + (1 - \mathcal{C}^A) \mathcal{C}^B \hat{\mathbf{N}}^A \cdot F_{D,C} + (1 - \mathcal{C}^B) \mathcal{C}^A \hat{\mathbf{N}}^A \cdot F_{C,D} + (1 - \mathcal{C}^A)(1 - \mathcal{C}^B) \hat{\mathbf{N}}^A \cdot F_{D,D} = 0 \\ \mathcal{C}^A \mathcal{C}^B \hat{\mathbf{N}}^B \cdot F_{C,C} + (1 - \mathcal{C}^A) \mathcal{C}^B \hat{\mathbf{N}}^B \cdot F_{D,C} + (1 - \mathcal{C}^B) \mathcal{C}^A \hat{\mathbf{N}}^B \cdot F_{C,D} + (1 - \mathcal{C}^A)(1 - \mathcal{C}^B) \hat{\mathbf{N}}^B \cdot F_{D,D} = 0 \end{cases} \quad (14)$$

This condition has the same intuition of the construction mentioned in Section 3.2: by appropriately selecting  $\mathcal{C}^A(\mathbf{Q})$  and  $\mathcal{C}^B(\mathbf{Q})$  one can obtain a vector field that is parallel to  $\Sigma^A$  and  $\Sigma^B$  first, and then bilinearity guarantees that it will be parallel to  $\Sigma$ . In this way, the trajectories of the system remain constrained on the double sliding surface as long as this is nodally attractive.

Because the symmetric system exhibits a pseudo-steady-state on the switching surface, we first look for one on the double sliding boundary in the asymmetric case. However, there exists no such pseudo-steady-state; since we argued that  $\bar{q}$  is such that

$$\hat{\mathbf{N}}^A \cdot F_{D,C}([\bar{q}, \bar{q}, x, y]^T) = 0,$$

it must be that the components parallel to the double sliding surface  $\Sigma$  of all vector fields

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<sup>36</sup>Omitting dependence on  $\mathbf{Q}$  for readability.

are strictly negative as long as the perceptions are above  $\bar{q}$ . But this then implies that there cannot exist any mixing of the type of [Equation \(14\)](#) such that the combined vector field on  $\Sigma$  vanishes. Formally, suppose Alice’s perceptions first reach  $\bar{q}$ : she is pushed into the region where she strictly prefers defecting, so the system leaves the 2-dimensional surface  $\Sigma$ .<sup>37</sup> Under the [Assumption S](#), the players switch actions exactly at the same time, and this allows for a pseudo-steady-state in which incentives are perfectly balanced. Instead, under asymmetry the players are allowed to switch at different times, and this generates more complex dynamics, which we study in the next section.

## B.2 Cyclic behavior

It may be worrisome that the asymmetric system lacks a pseudo-steady-state equivalent to the one of [Proposition 1](#), but we now argue that the long-term behavior of this four-dimensional differential inclusion is qualitatively similar both to what we observe in the simulations of [Section 3](#) and to the dynamics of spontaneous coupling. We start by looking at the trajectory of a solution to the differential inclusion. [Figure 10](#) represents the four-dimensional system in the two-dimensional space of differences between the agents perceptions, and we see immediately that there is no apparent resting point. However, the heatmap shows that most of the time is spent in a region where perceptions of cooperation and defection coincide. In other words, most of the time the system will exhibit Q-values equal for cooperation and defection for both Alice and Bob. This suggests that, while the trajectories do not remain on the  $\Sigma$  surface, they keep orbiting very close to it. Unfortunately, it is not possible to formally characterize this cyclic behavior. In fact, a closer inspection reveals that the trajectory behaves erratically and unpredictably, following a “butterfly” motion that resembles the well-known Lorentz attractor from chaos theory. We now verify that the four-dimensional differential inclusion without symmetry assumption is indeed a *chaotic* system, and that therefore we should *not* expect to find *any* steady state. For this, we need the following definition.

**Definition 9** (Informal). A dynamical system is said to be chaotic if the distance between two trajectories originating from different yet arbitrarily close starting points grows unboundedly large; for a formal statement in terms of Lyapunov exponents, see [Holden \(2014\)](#) and references therein.

Intuitively, a system is chaotic when its behavior, while deterministic, cannot be well predicted by its initial conditions. By definition, this can only happen for nonlinear dynamical systems, or when (even linear) systems exhibit discontinuities on the right-hand

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<sup>37</sup>Naturally, if it was Bob to reach  $\bar{q}$  first, he would be the first to start defecting most of the time.

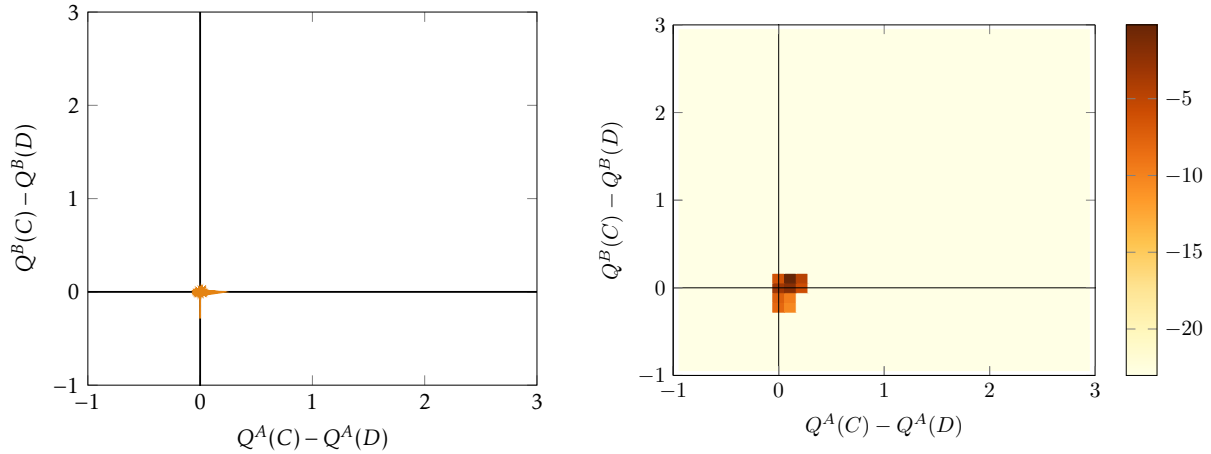


Figure 10: A chaotic trajectory of the system of Equation (4) in a 2-D representation. On the left, we depict the motion in the difference space. Agents appear chaotically attracted to a butterfly-like motion around the origin. The figure on the right represents the logarithmic density of time spent in a given square cell of side 0.2. Effectively, for over 90% of the time the algorithm’s perceptions are within  $\pm 0.5\%$  of each other.

side. Following the above definition of chaotic system, we plot in Figure 11 the evolution of  $Q_C^A(t)$  for two instances with slightly perturbed initial conditions,<sup>38</sup> as well as the distance between the two trajectories. Despite their very similar initial point the two trajectories vary wildly, and their distance increases by 20 orders of magnitude. This behavior further strengthens our intuition that we should not expect to find any steady state. It is then clear that, while restrictive, Assumption S is the only assumption that allows us to characterize formally the behavior we observe in simulations.

**Spontaneous coupling at play.** While the chaotic behavior of the solutions of the asymmetric system makes it impossible to prove stability results, we can still recover most of the intuitions obtained under Assumption S in Section 3. As we can see from Figure 10 above, even under asymmetric initialization each player’s perceptions remain close to each other most of the time; in other words, while the perceptions can (and will) remain different *across* players, they are almost always equal *within* each player. Looking at the description of spontaneous coupling we gave in Section 3.3, we see that for the cyclic behavior to happen it is not required that the perceptions be equal across players. In fact, it is sufficient that the perceptions of player  $i$  be close enough to each other, so that when the opponent switches to a different action also  $i$  quickly follows suit. We look for evidence of such correlated switches with Figure 12, which shows a measure of correlation

<sup>38</sup>The perturbation is of the order of  $e^{-20}$ .

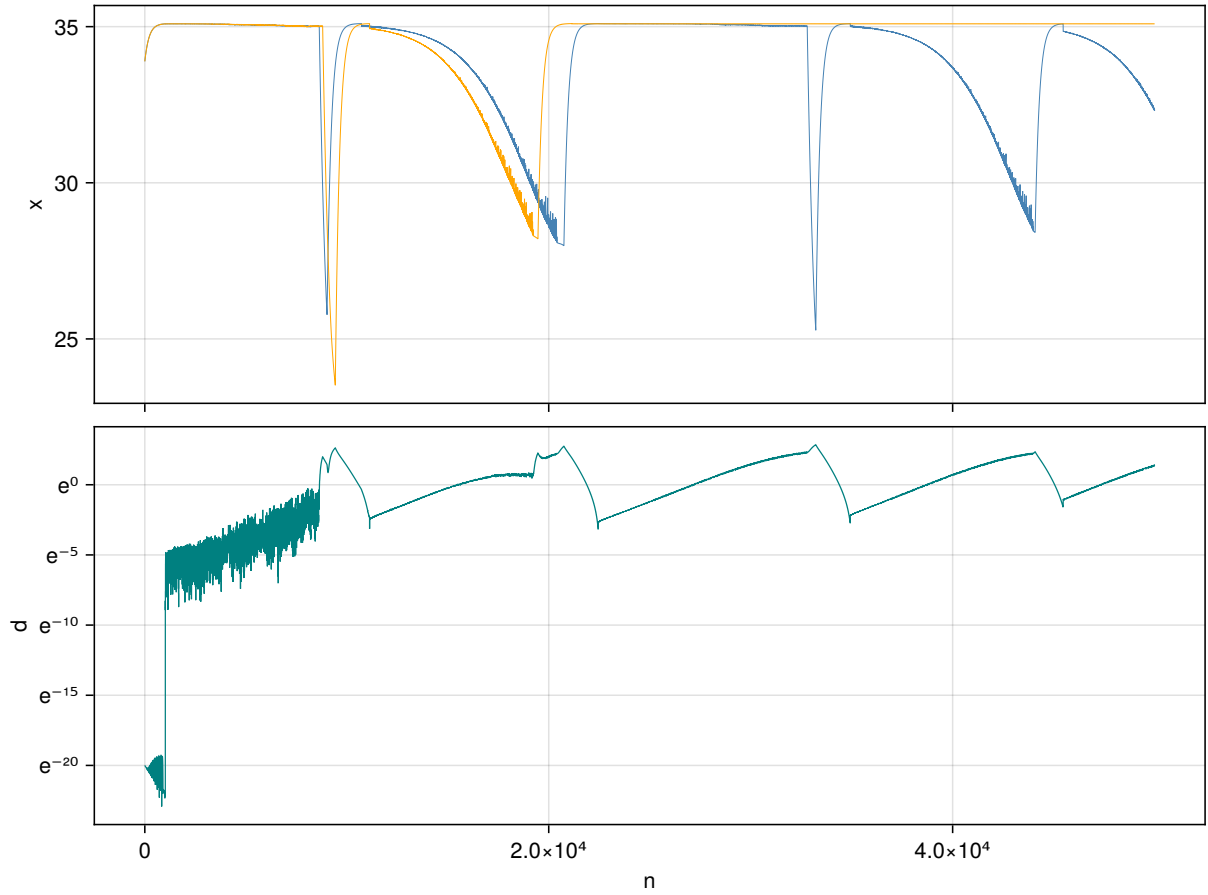


Figure 11: The top figure depicts the evolution of (the first component of) two trajectories with minimal perturbations of their initial conditions. The bottom figure represents the absolute distance between these two trajectories, in logarithmic scale. After an initial convergence, hitting the boundary leads to exponential divergence (the green line has constant slope upwards in the logarithmic scale). Finally the trajectories saturate, i.e. the boundedness of trajectories limits their absolute divergence.

in play over 500 different trajectories, each initialized at random, asymmetric starting points. In the figure we see that, whenever the value of cooperation is high enough to warrant spontaneous coupling, there is always positive and high correlation between the movements of the perceptions of the players. In other words, when one of them switches, the other quickly switches as well, and this sustains high level of cooperation as predicted by our theory. We interpret this as strong evidence that spontaneous coupling is at play also in the asymmetric case, despite the fact that perceptions can be different across players. Notice however that asymmetry is not immaterial, because it reduces the likelihood that spontaneous coupling, and therefore sustained cooperation, arise; however, the lack of tractability of the four-dimensional system prevents us from characterizing how less likely cooperation is. By comparing [Figure 13](#) with [Figure 2](#), which represent the same quantities under asymmetric and symmetric initialization respectively, we offer the following conjecture. The difference in the left panels of the figures suggests that asymmetry plays a role in determining which trajectories “embark” on a path of cooperation and which do not; clearly in [Figure 13](#) we see defection arising for value of  $g$  such that under symmetry cooperation is the only possible outcome. At the same time, asymmetry appears to affect “local time” (the amount of time that each player spends playing cooperate when they learn to cooperate) only marginally: each bar in [Figure 13](#) is lower than the corresponding in [Figure 2](#), but this difference is negligible for high values of  $g$ . Taken together, these observations further suggest that the mechanism whereby algorithms maintain coupling is the same in the symmetric and asymmetric case, and that the main difference is in the domain of initial values for which coupling can be reached.

The purpose of this appendix was to both justify the existence of [Assumption S](#) and to argue that imposing that assumption does not fundamentally alter the dynamics we aim to interpret. It is now clear that the assumption serves a dual purpose: in [Appendix B.1](#) we argued that assuming symmetry can be regarded as a way to circumvent the issue of non-uniqueness of a solution to the four-dimensional differential inclusion; we have now demonstrated that it is also a critical assumption to guarantee analytical tractability. At the same time, the numerical analysis above has shown that the theory of spontaneous coupling can be applied also to the system with asymmetric initialization and explain its behavior, although without symmetry we cannot characterize the “local time” that each player spends cooperating in the cycles. Overall, we believe that, while definitely not without loss, the assumption of symmetry is warranted to make a first step towards understanding the theory of algorithmic coordination; further research will be needed to improve our formal understanding of the asymmetric system.

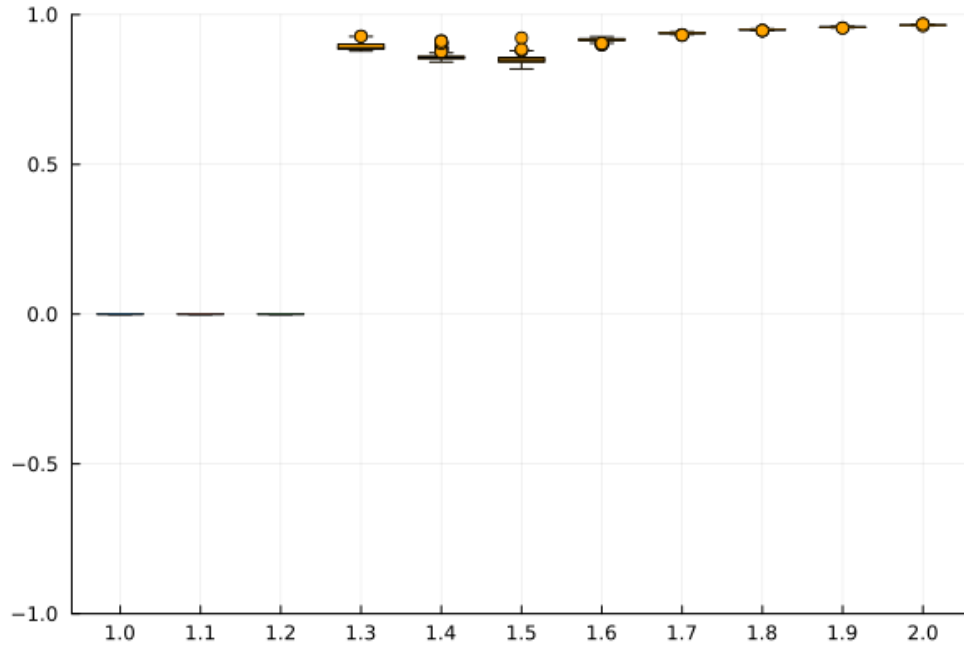


Figure 12: To obtain this plot we ran 500 simulations for each value of the cooperation parameter  $g$ , and for each simulation we computed a renormalized Simple Matching Coefficient for the actions taken by the algorithms in the last  $10^5$  periods. A value of  $-1$  means perfect anticorrelation and  $+1$  means perfect correlation. We show here the box plot of such simulations.

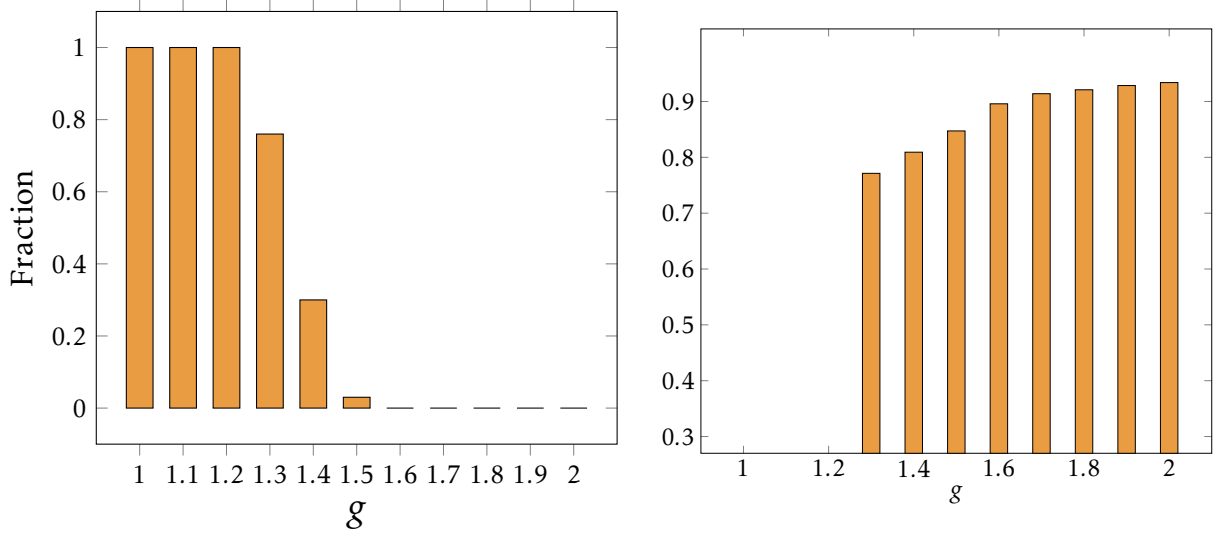


Figure 13: The left panel represents fraction of training runs where both players learn to play the Nash Equilibrium (as in Figure 2a). The right panel represents the ratio of periods where cooperation is a player's preferred action (as in Figure 2b) in a training run in spontaneous coupling sustains the cooperative outcome. Both panels are obtained by simulating algorithms with standard parametrizations but asymmetric initialization, sampled uniformly at random from an optimistic interval.

## Appendix C Game with $N$ players

In this Appendix we extend the model of Section 3 to a setting where the number of players is arbitrary. We first show that all our intuitions from Section 3.1 carry over to this setting. This richer setting also allows us to carry out comparative statics with respect to the number of players in the game: we uncover that spontaneous coupling becomes more difficult to sustain as the number of players increases, thus providing a theoretical justification for anecdotal and simulations-based evidence (e.g., Waltman and Kaymak, 2008) that when the number of players increases, and the actions of each of them affects payoffs in an increasingly negligible fashion, algorithmic collusion may disappear.

We now describe the stage game. Let  $N$  be the number of players in the game, each endowed with two dollars that can be contributed to a common pool; as before, the amount contributed increases by a factor of  $g$  and is then redistributed among all agents. Suppose that  $N_C \leq N$  contribute to the pool so that  $N_D = N - N_C$  do not. If player  $i$  contributes, her payoff is

$$r^i(C, N_C) = \frac{2N_C}{N}g,$$

while if she does not contribute (i.e., defects) her payoff is

$$r^i(D, N_C) = 2 + \frac{2N_C}{N}g.$$

To keep the analysis consistent with the previous sections, we assume here too that the players employ an  $\varepsilon$ -greedy Q-learning algorithm as described in [Section 3](#). We carry out the fluid approximation analysis under the same assumptions of [Section 3.1](#): in particular, we assume symmetry across players, both for the algorithm parameters and for initialization, so that the learning dynamics are a 2-D system of ODEs. Following [Section 3.1](#), the 2-D system gives the law of motion of the Q-values of a “representative” agent  $i$ . In what follows we will write

$$\begin{aligned}\omega_C &= \underbrace{\omega_{C, \dots, C}}_{N \text{ times}} = \left\{ Q \in \mathbb{R}^{2N} : Q_C^i > Q_D^i \quad \forall i = 1, \dots, N \right\} \\ \omega_D &= \underbrace{\omega_{D, \dots, D}}_{N \text{ times}} = \left\{ Q \in \mathbb{R}^{2N} : Q_C^i < Q_D^i \quad \forall i = 1, \dots, N \right\}\end{aligned}$$

That is, in  $\omega_C$  all players perceive that contributing is more profitable than not, and the opposite in  $D$ ; owing to the assumption of symmetry, the 2-D system only visits these two subsets of  $\mathbb{R}^{2N}$ .

Suppose that the system is in  $\omega_C$ : the number of agents playing action  $C$  other than the representative agent  $i$ , denoted by  $\tilde{N}_C$  is then a Binomial random variable with success probability equal to  $1 - \frac{\varepsilon}{2}$  (the probability that each agent plays  $C$ ) and number of trials equal to  $N - 1$ . It follows then that  $i$ 's expected payoff when playing  $C$  is

$$\mathbb{E}\left[r^i(C, \tilde{N}_C)\right] = \mathbb{E}\left[\frac{2\tilde{N}_C + 1}{N}g\right] = \frac{2g}{N}\left[(N-1)\left(1 - \frac{\varepsilon}{2}\right) + 1\right],$$

and when playing  $D$  it is instead

$$\mathbb{E}\left[r^i(D, \tilde{N}_C)\right] = \mathbb{E}\left[2 + \frac{2\tilde{N}_C}{N}g\right] = 2 + \frac{2g}{N}(N-1)\left(1 - \frac{\varepsilon}{2}\right).$$

It then follows that the vector field governing the dynamics in  $\omega_C$  is given by

$$\begin{cases} \dot{Q}_C = \alpha\left(1 - \frac{\varepsilon}{2}\right)\left[\frac{2g}{N}\left[(N-1)\left(1 - \frac{\varepsilon}{2}\right) + 1\right] + (\gamma - 1)Q_C\right] \\ \dot{Q}_D = \alpha\frac{\varepsilon}{2}\left[2 + \frac{2g}{N}(N-1)\left(1 - \frac{\varepsilon}{2}\right) - Q_D + \gamma Q_C\right] \end{cases} \quad (15)$$

If instead the system is in  $\omega_D$ , the number of agents other than  $i$  who contribute to the common pool is a Binomial random variable with success probability  $\frac{\varepsilon}{2}$  and number of trials equal to  $N - 1$ . Following the same steps as before we then obtain that the vector field describing the dynamics in  $\omega_D$  is

$$\begin{cases} \dot{Q}_C = \alpha \frac{\varepsilon}{2} \left[ \frac{2g}{N} \left[ (N-1) \frac{\varepsilon}{2} + 1 \right] - Q_C + \gamma Q_D \right] \\ \dot{Q}_D = \alpha \left( 1 - \frac{\varepsilon}{2} \right) \left[ 2 + \frac{2g}{N} (N-1) \frac{\varepsilon}{2} + (\gamma - 1) Q_D \right] \end{cases} \quad (16)$$

Because this is a 2-D system of ODEs with a single discontinuity surface, we can apply exactly the same theory we used in [Section 3.1](#); in particular, we refine [Proposition 1](#) to the following.

**Proposition 3.** *Let  $\tilde{\varepsilon}(g) = 1 - \sqrt{\frac{1}{g}}$  and  $\underline{\varepsilon}(g) = 1 - \sqrt{\frac{2-g}{g}}$ , and notice  $\tilde{\varepsilon}(g) \leq \underline{\varepsilon}(g)$ . The forward limit set of the ODE is always a singleton for any initial condition. If  $\varepsilon \geq \underline{\varepsilon}(g)$ , all initial conditions lead to the unique steady state*

$$q_D^{eq} = \left( \frac{N(\varepsilon g + 2) - \varepsilon g}{(1 - \gamma)N}, \frac{N(2\gamma + \varepsilon g) + 2(g - \varepsilon - g\gamma)}{(1 - \gamma)N} \right),$$

which lies in  $\omega_D$ .

If  $\tilde{\varepsilon}(g) < \varepsilon < \underline{\varepsilon}(g)$ , then there exists a threshold  $\underline{N}(g, \varepsilon)$  such that, if  $N \geq \underline{N}(g, \varepsilon)$  all initial conditions lead to the same, unique, steady state  $q_D^{eq}$ . Instead, if  $N < \underline{N}(g, \varepsilon)$  there exist two regions of attraction  $R_C$  and  $R_D$ . All initial conditions in  $R_D$  lead to the steady state  $q_D^{eq}$ . Initial conditions in  $R_C$  lead to the pseudo-steady-state (see [Definition 3](#))

$$q_C^{eq} = (y, y) \text{ where } y = \frac{N(1 + g) + \sqrt{(g-1)N[N(g-1) + g(\varepsilon-2)\varepsilon(N-1)]}}{(1 - \gamma)N},$$

which lies in  $\overline{\omega}_D \cap \overline{\omega}_C$ . The threshold is defined as

$$\underline{N}(g, \varepsilon) = \frac{\varepsilon^2 g - 2\varepsilon g}{g + \varepsilon^2 g - 2\varepsilon g - 1}.$$

Finally, if  $\varepsilon < \tilde{\varepsilon}(g)$ , then there always exist two regions of attractions  $R_D$  and  $R_C$ : all initial conditions in  $R_D$  lead to  $q_C^{eq}$ , and all initial conditions in  $R_C$  lead to  $q_C^{eq}$ .

The proof of this proposition follows verbatim along the lines of the proof of [Proposition 1](#) and is therefore omitted. [Proposition 3](#) generalizes [Proposition 1](#) by allowing an arbitrary number of players in the games, but its statement is familiar: for a given

value of the attractiveness of cooperation, if the exploration rate is too low a cooperative pseudo-steady-state is always present, while if it is high enough only defection remains as possible outcome. The main novelty with respect to [Proposition 1](#) is that there exists an “intermediate” regime, when  $\varepsilon \in [\underline{\varepsilon}(g), \bar{\varepsilon}(g)]$ , in which the number of players in the game determines whether spontaneous coupling is possible. When the exploration rate falls in this intermediate regime, the number of players cannot be too high for coupling to exist: the intuition is that as the number of players increases, the likelihood that at least one of them will defect while “cooperate” is the preferred action increases as well. However, this decreases the payoff that a player who is cooperating perceives in expectation, and we know from [Section 3.3](#) that spontaneous coupling hinges on the players perceiving a high payoff when they cooperate; therefore, when there are too many players, the expected payoff from cooperating is too low to sustain coupling, and the only possible steady state is the one where everybody defects.

The following corollary reinforces the intuition above: it shows that even when spontaneous coupling can exist, the proportion of time that players spend cooperating decreases in the number of agents. The local time described in [Corollary 4](#) is just the generalization to the  $N$  players setting of the object described in [Corollary 1](#) and interpreted in [Section 3.3](#).

**Corollary 4.** *In the pseudo-steady-state  $q_C^{eq}$  the agents spend a fraction  $\tau_c$  of their time cooperating, where*

$$\tau_c = \frac{1}{2} \left( 1 + \frac{\sqrt{(g-1)N(g((\varepsilon-2)\varepsilon(N-1)+N)-N)}}{(1-\varepsilon)(g-1)N} \right).$$

*Moreover,  $\tau_c$  is decreasing in the number of players.*

[Figure 14](#) gives a visual representation of [Corollary 4](#): the case  $N = 2$  clearly corresponds to [Figure 8b](#), but the bars from the simulation are omitted for better readability. As mentioned above, the local time  $\tau_c$  is decreasing in the number of players, and the decrease is sharper when cooperation is not as attractive. Moreover, [Figure 14](#) shows that the effect of increasing the number of players on the time spent cooperating is decreasing. This behavior suggests that the dynamics of spontaneous coupling are strong enough to maintain high rates of cooperation even with many players, and confirms that the reason why spontaneous coupling disappears when  $N$  grows too large is that the expected value of cooperating becomes too low, and not that players do not cooperate long enough.

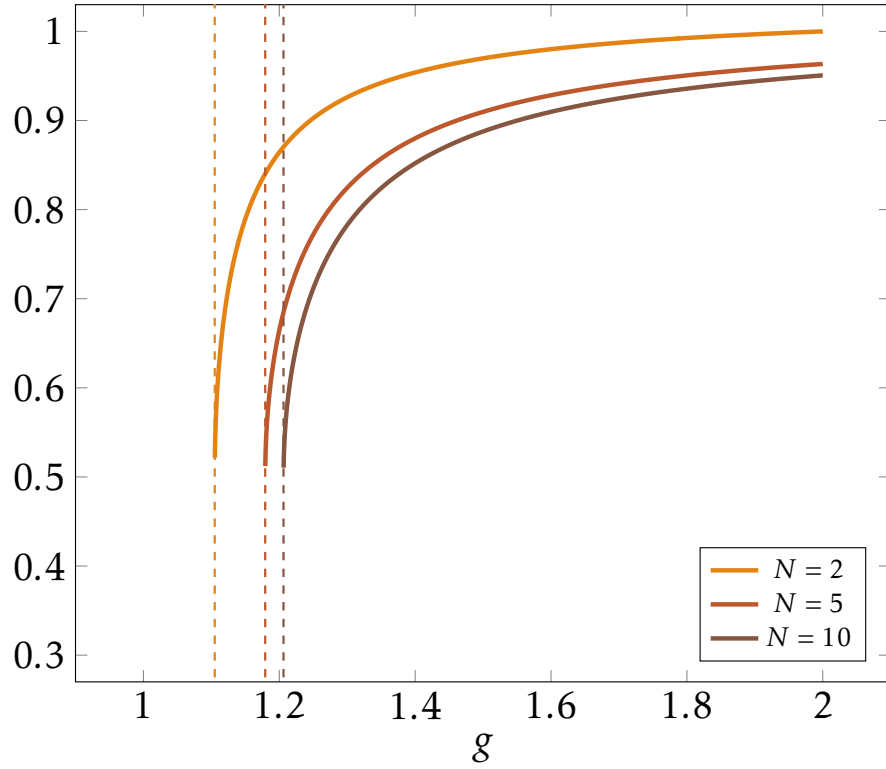


Figure 14: Proportion of time spent playing the cooperative action in  $q_C^{eq}$  according to [Corollary 4](#). The local time  $\tau_c$  is computed for three different number of players in the game, fixing the exploration rate at  $\varepsilon = 0.1$ . The dotted lines correspond to the lowest attractiveness of cooperation such that spontaneous coupling arises.

## Appendix D Market Division

We claimed in [Section 4](#) that spontaneous coupling may underpin a broad set of market manipulations by algorithmic agents. In support of that claim, in this section present a fully worked out version of the market splitting example we discussed after [Theorem 2](#). We show that spontaneous coupling can stifle competition by sustaining an anticompetitive conduct known as *market splitting*. With this conduct, market participants coordinate to concentrate each on a subset of the market and decline to participate in others, so that effectively each competitor is a monopolist in their reference market. We study this conduct in the context search advertising, where competing advertisers submit bids for keywords that have different values to them. We find that advertisers learn not to bid on their competitor’s favorite keyword, thus implicitly splitting the market. This outcome is supported by the familiar spontaneous coupling cycles

Consider a market where two advertisers, Alice.net and Bob.com, repeatedly compete in auctions for ad slots on three distinct keywords:  $a$ ,  $m$ , and  $b$ ; there is one slot available for each keyword. Each advertiser  $I$  derives a certain value from a click on their ad, denoted by  $v_j^I \sim U[1, 2]$ , where  $j \in \{a, m, b\}$  denotes the keyword. All values  $v_j^I$  are redrawn independently in every auction. We assume that the probability of a click varies depending on the keyword. Specifically, an ad on an advertiser’s “branded” keyword yields a higher click-through rate (CTR) than an ad on a neutral or their competitor’s branded keyword, which we model with advertiser-keyword specific click-through rates (CTR), denoted by  $ctr_j^I$  for advertiser  $I$  in keyword  $j$ . We choose  $ctr_a^A = 1$ ,  $ctr_m^A = 0.6$ , and  $ctr_b^A = 0.2$  for Alice.net, and symmetrically for Bob.com.<sup>39</sup>

The ad slots are allocated via (separate) second-price auctions with a reserve price of 1: Alice.net and Bob.com may submit their values  $v_j^I$  for all three keywords, and the highest bidder for the  $j$  auction wins the ad slot for the keyword  $j$ . The winner pays  $t_j^I$  per click, where  $t_j^I$  the second highest bid or the reserve price if they were the only bidder. The expected payoff for the winner is given by their CTR multiplied by the difference between their bid and their payment,  $ctr_j^I \times (v_j^I - t_j^I)$ . The loser receives a payoff of 0. If an advertiser does not submit a bid, they receive a payoff of 0. Notice that bidding on all three keywords is a dominant strategy in this game. This is because the payoff for winning a keyword is always positive, while the payoff for losing a keyword is 0, and there is no budget constraint.

Within the setting described, we let advertisers A and B use a learning algorithm to

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<sup>39</sup>Keywords denoted by  $a$  and  $b$  are “branded” for Alice.net and Bob.com, respectively. Think of a query that communicates a clear intent to reach Alice.net’s website. The query denoted by  $m$  instead is neutral — it may lead to a click on either advertiser with the same probability.

determine which keywords to bid on. Specifically, we assume that A and B implement a Q-learning algorithm that selects a subset of keywords to bid on. Note that the algorithm does not select a bid: we assume that when agents bid on a given keyword they bid their value. Thus, the action space for this algorithm is defined as the power set of the set of available keywords, which results in a total of  $2^3 = 8$  possible keyword combinations.

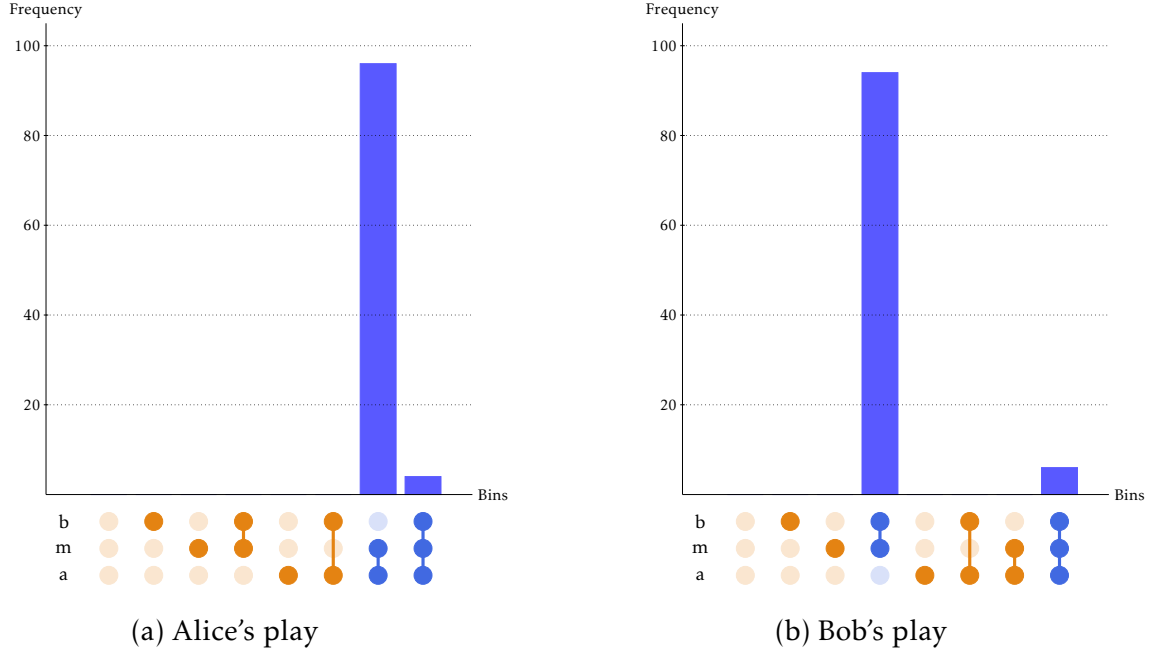


Figure 15: The figure shows empirical frequencies of the convergence outcomes of Alice.net and Bob.com's play. The left panel shows that Alice.net bids on all keywords with 64% probability, and similarly does Bob.com in the right panel. Instead, with 36% probability Alice.net and Bob.com *split the market*: both avoid bidding on their adversary's branded keyword. The frequencies are obtained by running 50 independent simulations with two independent Q-learning algorithms ( $\alpha = 0.005, \epsilon = 0.01, \gamma = 0.9$ , randomized optimistic, and symmetric initialization). We collect the actions the Q-learning algorithms converge to after  $10^6$  iterations.

To investigate the dynamics of the Q-learning algorithm, we simulate its behavior and visualize the results in Figures 15 and 16. In Figure 15, we observe that both algorithms exhibit convergence towards the dominant strategy in more than 50% of the independent, randomly initialized learning trajectories. However, in 18 out of 50 simulations, the two advertisers learn to collude by splitting the market. Under this collusion scheme, each advertiser only bids on his own branded keyword and the neutral one. In this collusive scheme, reminiscent of collusion in spatial models, neither advertiser bids on the opponent's branded keyword, despite the fact that it is strategically suboptimal. It is important to notice that the outcome of collusion by market splitting is Pareto dominant:

both players achieve better outcomes than if they bid on all keywords. In fact, bidding on all keywords yields an expected payoff of  $\frac{1}{6}\left(1 + \frac{6}{10} + \frac{2}{10}\right) = \frac{3}{10}$  per round, whereas market splitting yields an expected payoff of  $\frac{1}{2} + \frac{1}{6}\frac{6}{10} = \frac{6}{10}$  per round.

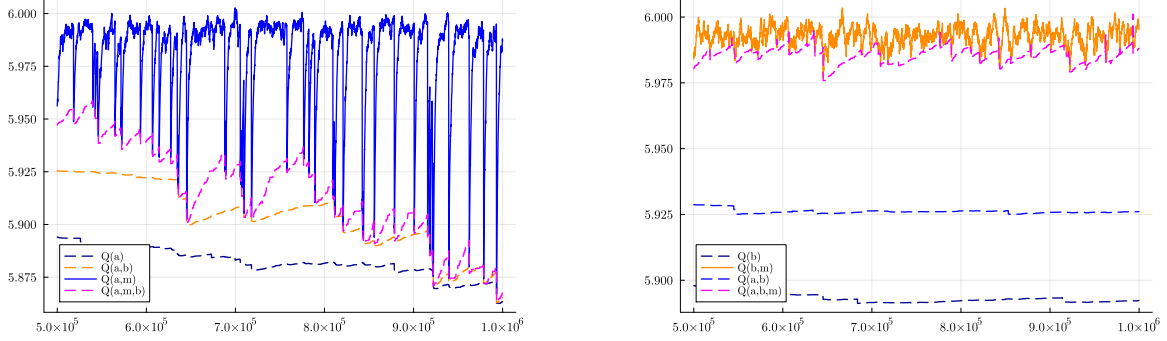


Figure 16: Learning trajectories from a simulation with  $\gamma = 0.9$ ,  $\varepsilon = 0.01$ ,  $\alpha = 0.005$ . The solid line represents  $Q(m, a)$  and  $Q(b, m)$  on the left and right, respectively. The dotted lines represent the  $Q$  values associated with some bundles containing the opponent's keyword.

We now focus on [Figure 16](#) to examine in greater detail the collusive dynamics. This figure portrays the evolution of the Q-learning algorithm for a subset of actions taken by each advertiser, and it is taken from one of the experiments where the advertisers learn to split the market. As the figures reveal, the collusion behavior is not perfect. Although most of the time the two advertisers engage in collusive splitting, experimentation eventually brings about the realization that they could benefit by bidding on the opponent's branded keyword. As soon as they realize this, one of the advertisers begins bidding on the opponent's branded keyword, causing a sudden drop in the opponent's payoffs, which leads her to abandon the splitting strategy and revert to bidding on all keywords. This sequence of events causes both players to experience significantly lower payoffs, but their expectations (their  $Q$ -value) about the market-splitting outcome persistently remain high. Ultimately, both advertisers return to splitting the market simultaneously in a stochastic recurrent cycle typical of spontaneous coupling, as described in [Section 3.3](#).

Recall that in the keyword advertising auction both algorithms have access to eight actions each. Additionally, their payoffs are stochastic, because the value for each keyword is drawn in every period from a uniform distribution. In this context, we would expect the statistical linkage to be weaker: noisy observations and large number of actions should intuitively prevent coordination. Nevertheless, we observe spontaneous coupling with considerable frequency. We interpret this as additional evidence that a system of independent algorithms can endogenously form coordination incentives.

## Appendix E Learning-robust mechanism design

In this Appendix we formalize the somewhat loose definitions given in [Section 5](#) and we prove [Theorem 5](#).

First, note that we can define an equivalence relation on the outcome space,  $\sim_i$ , such that  $x \sim_i y$  if and only if  $u^i(x, \lambda^i) = u^i(y, \lambda^i)$  for all  $\lambda^i \in \Lambda_i$ . Let  $\mathcal{X}_i = \mathcal{X} / \sim_i$  be the quotient of the outcome space with respect to this equivalence relation. We will refer to an element of  $\mathcal{X}_i$ , an equivalence class  $[x]_i$ , as an  $i$ -outcome.

Let us define formally what it means to provide ex-post feedback.

**Definition 10.** A *feedback policy* for mechanism  $f$  and agent  $i$  is a factorization of  $f$  through a partition. It is composed of the following elements:

- A *signal space*  $S$ , which is a partition of  $\Lambda_{-i}$ ;
- A map  $\phi_i: \Lambda_{-i} \rightarrow S$ ;
- A map  $g: \Lambda_i \times S \rightarrow \mathcal{X}_i$ ;

such that

- The diagram commutes, i.e.  $[f(\lambda^i, \lambda^{-i})]_i = g(\lambda^i, \phi_i(\lambda^{-i}))$  for all  $\lambda^i, \lambda^{-i}$ ;
- The map  $\phi_i$  is a partition map, i.e.  $\phi_i(\lambda^i) \ni \lambda^i$ .

We denote a feedback policy by its map  $\phi_i$ . A collection  $(\phi_i)_i$  of feedback policies for each agent  $i$  is a *feedback structure*.

Remember the loose definition of [Section 5](#): a feedback policy for agent  $i$  is a partition of the space of opponents' types,  $\Lambda_{-i}$ . This partition is such that agent  $i$  can evaluate what outcome he could have enforced had he unilaterally deviated to a different report  $\lambda^i$ . We request that a feedback policy allows agent  $i$  to compute all of his  $i$ -outcomes for any given report.

Next, we formalize the desire for reduced communication and revelation. We define the following partial order on feedback policies:

**Definition 11.** Feedback policy  $\phi_i$  is *more private* than feedback policy  $\psi_i$ , denoted  $\phi_i \succeq \psi_i$ , if  $\phi_i(\lambda^{-i}) \supseteq \psi_i(\lambda^{-i})$  for all  $\lambda^{-i}$ .

Any two type profiles that are indistinguishable under a less private rule should be indistinguishable under a more private rule. As mentioned, the privacy order is a weak

partial order: not all feedback policies are comparable. However, it turns out that under the privacy order maximum and minimum are well-defined: the feedback policies together with the privacy order form a lattice.

**Proposition 4.** *Feedback policies together with the privacy order form a complete lattice.*

*Proof.* We simply need to show that for any two elements  $\phi_i, \psi_i$  there exist a join  $\phi_i \vee \psi_i$  and a meet  $\phi_i \wedge \psi_i$  which satisfy the feedback policy definition. The argument follows from the lattice structure of the set of partitions with the partial order *coarser-than*. Note that  $\phi_i^{-1}(\{x: x \in \Lambda_{-i}\})$  defines a partition of the space  $\Lambda_{-i}$ , and the same is true for the  $\psi_i$ . We then require the preimage of join  $(\phi_i \vee \psi_i)$  to be the finest partition which is coarser than both the preimages of  $\phi_i$  and  $\psi_i$ . Formally, let  $A \subset \phi_i^{-1}(\{x: x \in \Lambda_{-i}\}) \vee_P \psi_i^{-1}(\{x: x \in \Lambda_{-i}\})$ , then

$$(\phi_i \vee \psi_i)(\lambda^{-i}) = (\phi_i \vee \psi_i)(\hat{\lambda}^{-i}) \text{ for all } \lambda^{-i}, \hat{\lambda}^{-i} \in A$$

Similarly, define the meet as the function  $\phi_i \wedge \psi_i$  such that it is constant over the meet of the two partitions. Again, let  $B \subset \phi_i^{-1}(\{x: x \in \Lambda_{-i}\}) \wedge_P \psi_i^{-1}(\{x: x \in \Lambda_{-i}\})$ , then

$$(\phi_i \wedge \psi_i)(\lambda^{-i}) = (\phi_i \wedge \psi_i)(\hat{\lambda}^{-i}) \text{ for all } \lambda^{-i}, \hat{\lambda}^{-i} \in B$$

The completeness of the lattice structure descends directly from the completeness of the partition lattice.  $\square$

Proposition 4 implies that there exist both a minimally- and a maximally-private feedback policy. The minimally-private policy is the full-revelation feedback policy: it reveals all information, and it is clearly less private than any other feedback policy. The maximally-private rule instead is a menu description.<sup>40</sup>

**Definition 12.** Let  $[x]_i$  be the equivalence class of outcome  $x$  in  $\mathcal{X}_i$ . A *menu* for mechanism  $f$  and agent  $i$ , given reports  $\lambda^{-i}$  of the opponents, is the set

$$\mathcal{M}_{\lambda^{-i}} = \left\{ [f(\hat{\lambda}^i, \lambda^{-i})]_i \mid \hat{\lambda}^i \in \Lambda_i \right\}.$$

That is, the menu is the set of outcomes such that agent  $i$  could have received had he reported any type in his type space.

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<sup>40</sup>Note that we defined the privacy lattice for feedback policies, not for feedback structures. When we analyze feedback structures, we implicitly consider the product lattice on the product space of feedback policies. This is correct because we assume private communication, but the analysis would likely change if we allowed public communication.

We can show that from the collection of all possible menus  $\{\mathcal{M}_{\lambda^{-i}}\}_{\lambda^{-i}}$  we can construct a feedback policy, and it is the maximally private feedback policy for agent  $i$ .

**Lemma 3.** *There exists a feedback policy  $\mu^i$  corresponding to the collection of menus  $\{\mathcal{M}_{\lambda^{-i}}\}_{\lambda^{-i}}$ , and  $\mu^i$  is the maximally private feedback policy for agent  $i$ .*

*Proof.* Consider the space  $\Lambda_{-i}$  with the following equivalence relation:

$$\lambda^{-i} \sim \hat{\lambda}^{-i} \text{ iff } \mathcal{M}_{\lambda^{-i}} = \mathcal{M}_{\hat{\lambda}^{-i}}$$

and denote its quotient by  $\Lambda_{-i}/\sim$ . An element of the quotient is an equivalence class of opponents' types, denoted by  $[\lambda^{-i}]$ . Since  $\sim$  is an equivalence relation, the quotient set  $\Lambda_{-i}/\sim$  is a partition of  $\Lambda_{-i}$ . Let

$$\begin{aligned} \mu^i: \Lambda_{-i} &\rightarrow \Lambda_{-i}/\sim \\ \lambda^{-i} &\mapsto [\lambda^{-i}] \end{aligned}$$

Let us show first that  $\mu^i$  satisfies the definition of feedback policy. Of course,  $\mu^i$  is a partition map: an element always belongs to its equivalence class. Define  $g$  as the function  $g(\lambda^i, [\lambda^{-i}]) := [f(\lambda^i, \lambda^{-i})]_i$ . Commutativity then follows from the fact that for all  $\lambda^{-i} \in [\lambda^{-i}]$  the menus  $\mathcal{M}_{\lambda^{-i}}$  coincide.

Now, we need to show that there is no feedback policy which is more private than  $\mu^i$ . Equivalently, we can show that there is no feedback policy  $\phi^i$  such that  $\phi^i \triangleright \mu^i$ . Suppose instead such a  $\phi^i$  exists. Then, it must be that there exists a  $\lambda^{-i}$  such that  $\phi^i(\lambda^{-i}) \supset \mu^i(\lambda^{-i})$ . Then there exists a  $\hat{\lambda}^{-i} \in \phi(\lambda^{-i})$  such that  $\hat{\lambda}^{-i}$  is not in the same equivalence class of  $\lambda^{-i}$ . This implies that  $\mathcal{M}_{\lambda^{-i}}$  is a different menu than  $\mathcal{M}_{\hat{\lambda}^{-i}}$ . Then, there exists a  $\lambda^i$  such that  $[f(\lambda^i, \lambda^{-i})]_i \neq [f(\lambda^i, \hat{\lambda}^{-i})]_i$ . We have then that  $g(\lambda^i, \phi(\lambda^{-i})) = [f(\lambda^i, \lambda^{-i})]_i \neq [f(\lambda^i, \hat{\lambda}^{-i})]_i = g(\lambda^i, \phi(\lambda^{-i}))$ , a contradiction.  $\square$

We observe another interesting property of the privacy order that stems from its connection with the set of partitions of the power set of  $\Lambda_{-i}$ :

**Corollary 5.** *The communication complexity of a feedback policy is monotonically decreasing in the privacy order.*

The maximally private feedback policy has the lowest communication complexity and it maintains the highest level of privacy. Our characterization says that the most private mechanisms are not particularly esoteric: they provide menu descriptions, which are well-known for their simplicity.